Hyperbola

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Hyperbola

The fixed point is called the focus, the fixed line is called the directrix and the constant ratio is called the eccentricity of the hyperbola and denoted by e.

In the given figure, S is the focus and N´N is the directrix.

Let P be any point on the hyperbola, then $\frac{PS}{PM} = e, e > 1$. Equation of a hyperbola can be obtained if the coordinates of its focus, equation of its directrix and eccentricity are given.

Definition

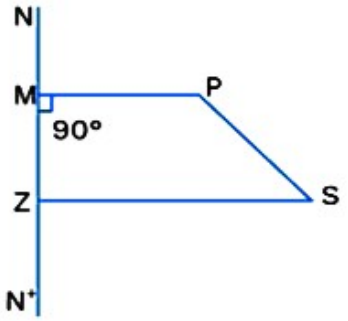
A hyperbola is the locus of a point which moves in a plane such that the ratio of its distance from a fixed point and a given straight line is always constant.

STANDARD EQUATION OF A HYPERBOLA

Let S be the focus and ZN is the directrix of an hyperbola. Draw perpendicular from S to the directrix which meet it at Z. A moving point is on the hyperbola such that

PS = e.PM

then there is point lies on the line SZ and which divide SZ internally at A and externally at A' in the ratio of e:1.



Therefore SA = e.AZ ...(i)

SA' = e.A'Z ...(ii)

Let AA' = 2a & take C as mid point of AA'

$\therefore$ CA = CA' = a

Add (i) and (ii)

$SA + SA' = e(AZ + A'Z)$

$(CS - CA) + (CA' + CS) = e[CA - CZ + CA' + CZ]$

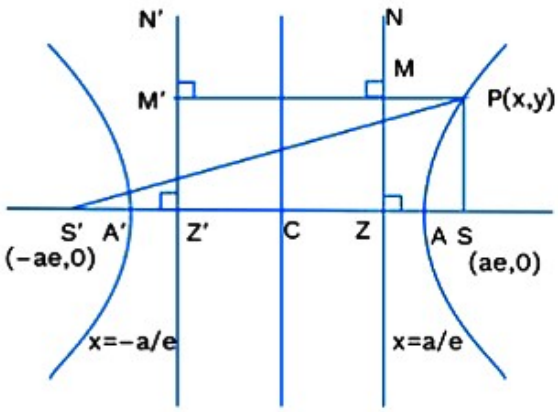
CS = ae

Subtract (ii) & (i), we get

SA'–SA = e(A'Z – AZ)

$(CA' + CS) - (CS - CA) = e[(CA' + CZ) - (CA - CZ)]$

$2CA = 2e \cdot CZ \Rightarrow CZ = \frac{a}{e}$



Consider CZ line as x -axis, C as origin & perpendicular to this line and passes through C is considered as y-axis. Now represent important parameters on coordinate plane. Let P(x, y) is a moving point, then
By definition of hyperbola.

$PS = ePM \Rightarrow (PS)^2 = e^2(PM)^2$

$\Rightarrow (x-ae)^2 + (y-0)^2 = e^2\left(x-\frac{a}{e}\right)^2 \Rightarrow (x-ae)^2 + y^2 = (a-ex)^2$

$\Rightarrow x^2 + a^2e^2 - 2xae + y^2 = a^2 + e^2x^2 - 2xae \Rightarrow x^2\left(1-e^2\right) + y^2 = a^2\left(1-e^2\right)$

$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{a^2\left(1-e^2\right)} = 1$ or $\frac{x^2}{a^2} - \frac{y^2}{a^2\left(e^2-1\right)} = 1$ where $b^2 = a^2\left(e^2-1\right)$

Hence equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $b^2 = a^2\left(e^2-1\right)$

TRACING OF HYPERBOLA

Equation of given hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ ...(i)

(i) If we put y = 0, then we see that hyperbola cuts x -axis at (±a, 0)

(ii) If we put x = 0, then $y^2 = -b^2$. Hence hyperbola does not cut y-axis.

(iii) When y is replaced by –y, the equation of hyperbola does not change, hence hyperbola is symmetric about x-axis. (Since equation contain even power of y therefore curve will be symmetric about x -axis).

(iv) When x is replaced by –x, then equation of hyperbola does not change, hence hyperbola is symmetric about y-axis. (Since equation contain even power of x therefore curve will be symmetric about y-axis).

(v) From (1), $y = \pm\frac{b}{a}\sqrt{x^2-a^2}$, since y is real

$\Rightarrow x^2 - a^2 \geq 0$ or $x \in (-\infty, -a] \cup [a, \infty)$

For each $x \geq a$ or $x \leq -a$ there are two values of y symmetrically situated on both side of x -axis.

Hence, the curve denoted by (1) consist of two symmetrical branches, each extending to infinite in two direction.

FACTS ABOUT THE HYPERBOLA

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

(i) By symmetry of equation of hyperbola, if we take second focus $(-ae, 0)$ and second directrix $x = -\frac{a}{e}$ and perform same calculation then we get same equation of hyperbola. This suggest that their two foci are $(ae, 0)$ and $(-ae, 0)$ and the two corresponding directrices as $x = \frac{a}{e}$ and $x = -\frac{a}{e}$. If focus is $(ae,0)$, then corresponding directrix is $x = \frac{a}{e}$ and if focus is $(-ae, 0)$, then corresponding directrix is $x = -\frac{a}{e}$.

(ii) By definition of hyperbola, the distance of any point P on the hyperbola from focus =e.(the distance of P from the corresponding directrix)

(iii) Distance between foci SS'=2ae distance between directrix $ZZ' = 2\frac{a}{e}$.

(iv) Two hyperbola are said to be similar if they have same eccentricity.

(v) The hyperbola has two branches neither of them cut the y-axis (conjugate axis).

(vi) Since the fundamental equation to the hyperbola only differs from that to the ellipse in having $-b^2$ instead of b^2. It is observed that many propositions for the hyperbola are derived from those for the ellipse by simply changing the sign of b^2.

(vii) Eccentricity $e = \sqrt{1 + \frac{b^2}{a^2}}$. Also, $b^2 = a^2\left(e^2 - 1\right)$

The smaller the e, the smaller will be the value of b for a given a. Therefore, as e decreases for a given a, the branches of the hyperbola would be bending towards x-axis. As e increases, the branches open up.

(viii) Equation of hyperbola when its transverse and conjugate axes are x and y-axes respectively is

$$\frac{x^2}{a^2} - \frac{y^2}{a^2\left(e^2 - 1\right)} = 1 \text{ or } x^2 - \frac{y^2}{\left(e^2 - 1\right)} = a^2$$

If e kept constant and $a \to 0$, then hyperbola will tend to pair of straight lines $x^2 - \frac{y^2}{\left(e^2 - 1\right)} = 0$, both passing through the origin.

Thus, in the situated limiting case of hyperbola is a pair of straight lines.

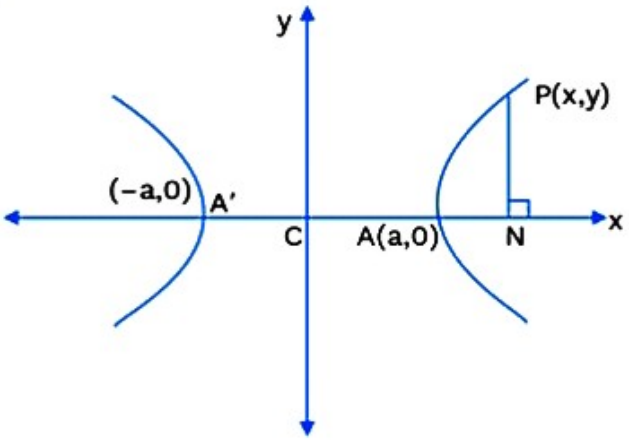
(ix) Since $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\therefore \quad \frac{y^2}{b^2} = \frac{x^2}{a^2} - 1 = \frac{(x-a)(x+a)}{a^2}$$

From figure,

$AN = CN - CA = x - a$

$A'N = CN + CA = x + a$ and $PN = y$



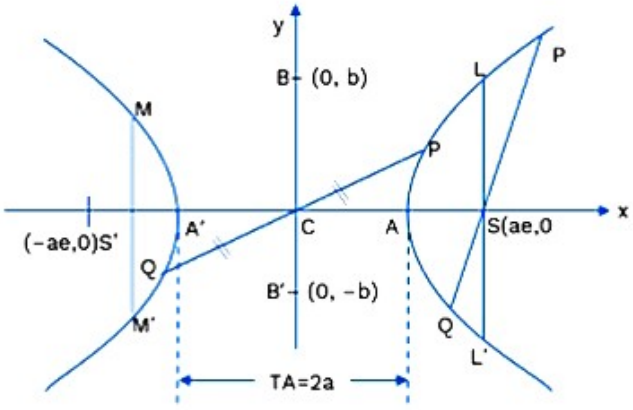
TERMS RELATED TO HYPERBOLA

(1) Centre :

In the figure, C is the centre of the hyperbola. All chords passing through C are called diameter and bisected at C.

(2) Foci :

S(ae, 0) and S'(−ae, 0) are two foci of hyperbola. Line containing the fixed points S and S' (called Foci) is called Transverse Axis (TA) or Focal Axis and the distance between S and S' is called Focal Length.



(3) Axes :

The line AA' is called transverse axis and the line BB' is perpendicular to it and passes through the centre (0,0) of the hyperbola is called conjugate axis.

The points of intersection of the directrix with the transverse axis are known as Foot of the directrix (Z and Z').

(4) Vertex:

The points of intersection (A, A') of the curve with the transverse axis are called Vertices of the hyperbola.

Point to Remember!!!

The length of transverse and conjugate axes are taken as 2a and 2b respectively.

The transverse and conjugate axes together are called principal axes of hyperbola and their intersection point is called the centre of hyperbola.

(5) Double ordinate:

Any chord perpendicular to the Transverse axis is called a Double Ordinate.

(6) Latus-rectum:

When double ordinates passes through the focus of parabola then it is called the latus rectum. In the given figure LL' and MM' are the latus-rectums of the hyperbola.

let LL' = 2k then LS = L'S = k

Let L(ae, k) lie on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$\therefore \quad \frac{a^2e^2}{a^2} - \frac{k^2}{b^2} = 1$

Or

$$k^2 = b^2\left(e^2 - 1\right) = b^2\left(\frac{b^2}{a^2}\right) \quad \left[\because b^2 = a^2\left(e^2 - 1\right)\right]$$

$$\therefore \quad k = \frac{b^2}{a} \quad (\because k > 0)$$

$$\therefore \quad 2k = \frac{2b^2}{a} = LL'$$

Length of latus rectum

$$= LL' = \frac{2b^2}{a} = \frac{(CA)^2}{TA} = 2a\left(e^2 - 1\right) = 2e\left(ae - \frac{a}{e}\right)$$

= (2e). (distance between the focus and the foot of the corresponding directrix)

End points of latus-rectums are

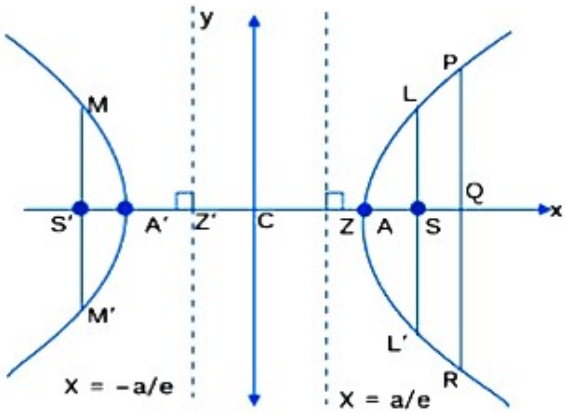
$$L = \left(ae, \frac{b^2}{a}\right), L' = \left(ae, -\frac{b^2}{a}\right);$$

$$M = \left(-ae, \frac{b^2}{a}\right), M' = \left(-ae, -\frac{b^2}{a}\right)$$

respectively.

(7) Focal chord:

A chord of hyperbola passing through its focus is called a focal chord.



Point to Remember!!!

ECCENTRICITY

For the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ we have $b^2 = a^2\left(e^2 - 1\right)$

$$\Rightarrow e = \sqrt{1 + \left(\frac{b^2}{a^2}\right)}$$

$$\Rightarrow e = \sqrt{\left\{1 + \frac{(\text{ conjugate axis })^2}{(\text{ transverse axis })^2}\right\}}$$

Eccentricity defines the curvature of the hyperbola and is mathematically spelled as :

$$e = \frac{\text{distance from centre to focus}}{\text{distance from centre to vertex}}$$

PARAMETRIC EQUATIONS OF THE HYPERBOLA

Let $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ be the hyperbola with centre C and transverse axis A'A.

Therefore circle drawn with centre C and segment A'A as a diameter is called auxiliary circle of the hyperbola.

$\therefore$ Equation of the auxiliary circle is $x^2 + y^2 = a^2$

Let P(x,y) be any point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Draw PN perpendicular to x-axis.

Let NQ be a tangent to the auxiliary circle $x^2 + y^2 = a^2$ Join CQ and let

$\angle QCN = \phi$ then P and Q are the corresponding points of the hyperbola and the auxiliary circle. Here ϕ is the eccentric angle of P. $(0 \le \phi < 2\pi)$.

Since $Q = (a\cos\phi, a\sin\phi)$

Now $x = CN = CQ\sec\phi = \sec\phi \cdot a$

$\therefore$ $P(x, y) = (a\sec\phi, y)$

$\because$ P lies on $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

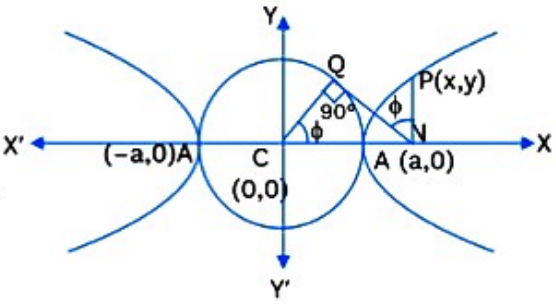
$\therefore$ $\frac{a^2\sec^2\phi}{a^2} - \frac{y^2}{b^2} = 1$ or $\frac{y^2}{b^2} = \sec^2\phi - 1 = \tan^2\phi$

$\therefore$ $y = \pm b\tan\phi$

$\therefore$ $y = b\tan\phi$ (P lies in 1st quadrant)

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Position of points Q an auxiliary circle and corresponding point P which describes the hyperbola or shown below in the table. Here $0 \le \phi < 2\pi$.



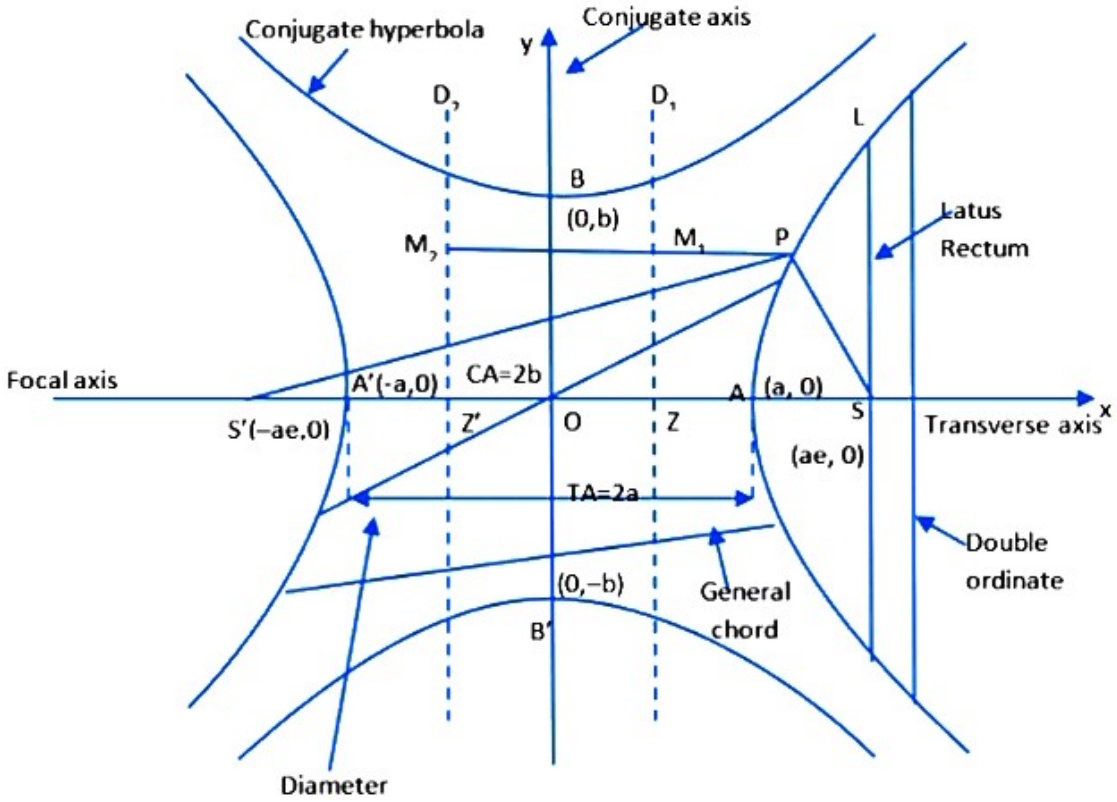
Point to Remember!!!

The equations of $x = a\sec\phi$ and $y = \tan\phi$ are known as the parametric equations of the hyperbola.

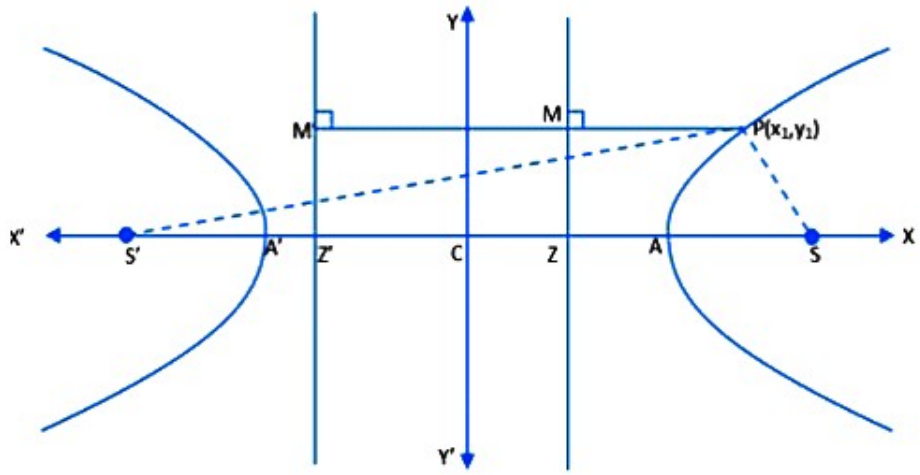
ϕ varies from	Q(a cosϕ, a sinϕ)	P(a secϕ , b tanϕ)
0 to $\pi/2$	II	III
$\pi/2$ to π	II	III
π to $3\pi/2$	III	II
$3\pi/2$ to 2π	IV	IV

HYPERBOLA AT A GLANCE

Parametric coordinates $x = a\sec\theta$ and $y = b\tan\theta$



FOCAL DISTANCE OF A POINT ON HYPERBOLA



The hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. The foci S and S' are (ae,0) and (−ae, 0) corresponding directrix are $x = \frac{a}{e}$ and $x = -\frac{a}{e}$ and respectively.

Let $P(x_1, y_1)$ be any point on (1).

Now $SP = ePM = e\left(x_1 - \frac{a}{e}\right) = ex_1 - a$

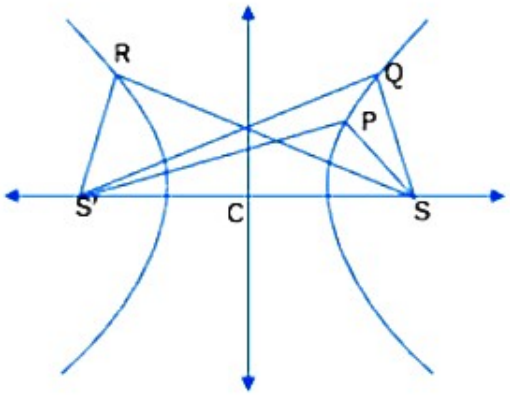
And $S'P = ePM' = e\left(x_1 + \frac{a}{e}\right) = ex_1 + a$

$\therefore \quad S'P - SP = (ex_1 + a) - (ex_1 - a) = 2a$

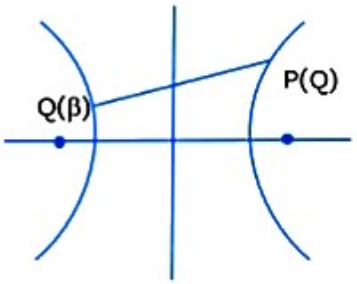
= AA' Transverse axis

Thus hyperbola is the locus of a point which moves in a plane such that the difference of its distances from two fixed point (foci) is constant and always equal to transverse axis. Hence, in the given figure length of transverse axis is given by

$PS' - PS = QS' - QS = RS - RS'$



CHORD OF HYPERBOLA



Chord joining two points with eccentric angles α and β is given by

$$\frac{x}{a}\cos\frac{\alpha-\beta}{2}-\frac{y}{b}\sin\frac{\alpha+\beta}{2}=\cos\frac{\alpha+\beta}{2} \quad ...(1)$$

if (1) passes through (d, 0) then

$$\frac{d}{a}\cos\frac{\alpha-\beta}{2}=\cos\frac{\alpha+\beta}{2}$$

$$\frac{d}{a}=\frac{\cos\frac{\alpha}{2}\cos\frac{\beta}{2}-\sin\frac{\alpha}{2}\sin\frac{\beta}{2}}{\cos\frac{\alpha}{2}\cos\frac{\beta}{2}+\sin\frac{\alpha}{2}\sin\frac{\beta}{2}}$$

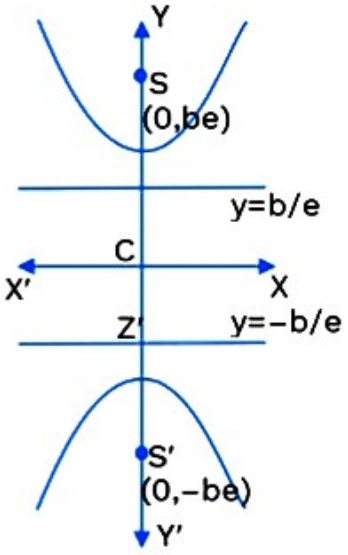
$$\frac{d+a}{d-a}=-\frac{\cos\frac{\alpha}{2}\cos\frac{\beta}{2}}{\sin\frac{\alpha}{2}\sin\frac{\beta}{2}}\Rightarrow\tan\frac{\alpha}{2}\tan\frac{\beta}{2}=\frac{a-d}{a+d}$$

If it passes through focus (ae, 0) then

$$\tan\frac{\alpha}{2}\cdot\tan\frac{\beta}{2}=\frac{1-e}{1+e}$$

CONJUGATE HYPERBOLA

Corresponding to every hyperbola there exist a hyperbola such that, the conjugate axis and transverse axis of one is equal to the transverse axis and conjugate axis of other, such hyperbolas are known as conjugate to each other.



Hence for the hyperbola, $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ ...(1)

the conjugate hyperbola is, $\frac{x^2}{a^2}-\frac{y^2}{b^2}=-1$ ...(2)

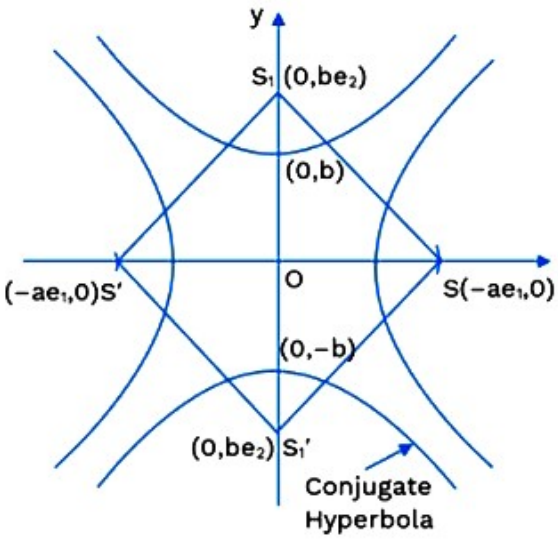
Comparison between hyperbola and its conjugate hyperbola

Centre	(0,0)	(0,0)
Length of transverse axis	2a	2b
Length of conjugate axis	2b	2a
Eccentricity	$b^2 = a^2(e^2 - 1)$, $e = \sqrt{1 + \frac{b^2}{a^2}}$	$a^2 = b^2(e^2 - 1)$, $e = \sqrt{1 + \frac{a^2}{b^2}}$
Foci	$(\pm ae, 0)$	$(0, \pm be)$
Equation of directrix	$x = \pm \frac{a}{e}$	$y = \pm \frac{b}{e}$
Length of latus rectum	$\frac{2b^2}{a}$	$\frac{2a^2}{b}$
Parametric coordinate	$(a\sec\phi, b\tan\phi)$ $0 \le \phi < 2\pi$	$(a\tan\phi, b\sec\phi)$ $0 \le \phi < 2\pi$
Focal distances	$ex_1 \pm a$	$ey_1 \pm b$
Difference of focal distances	2a	2b
Equation of transverse axis	y = 0	x = 0
Equation of conjugates	x = 0	y = 0
Tangent at vertices	$x = \pm a$	$y = \pm b$

Note:

(i) The foci of a hyperbola and its conjugate are concyclic and form the vertices of a square.

(ii) If e_1 and e_2 are the eccentricities of the hyperbola & its conjugate then $e_1^{-2} + e_2^{-2} = 1$



RECTANGULAR HYPERBOLA

If the lengths of transverse and conjugate axes of any hyperbola be equal then it is called rectangular or equilateral hyperbola. Since length of transverse axis and conjugate axis are same i.e. a = b.

Then $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ becomes $x^2 - y^2 = a^2$

Eccentricity $e = \sqrt{\left(1 + \frac{b^2}{a^2}\right)} = \sqrt{2}$

All the results of $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ are applicable to the hyperbola $x^2 - y^2 = a^2$ after changing b by a.

FIND ALL THE PARAMETERS OF A HYPERBOLA

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$

This equation is the form of $\frac{X^2}{a^2} - \frac{Y^2}{b^2} = 1$,

where X = x − h and Y = y − k

(i) Length of semi-transverse axis =a, length of semi-conjugate axis =b

(ii) Equation of transverse axis is Y =0 i.e. y − k =0
Equation of conjugate axis is X = 0, i.e. x − h = 0

(iii) Coordinates of centre is given by X=0 and Y=0 i.e., x − h = 0 and y − k =0. Therefore, centre is (h, k)

(iv) Eccentricity of the hyperbola $e = \sqrt{1+\frac{b^2}{a^2}}$

(v) Coordinates of vertices of the hyperbola are given by $X = \pm a, Y = 0$ i.e., $x - h = \pm a, y - k = 0$.
Hence vertices are $(h \pm a, k)$

(vi) Coordinate of foci are given by $X = \pm ae, Y = 0$
i.e., $x-h = \pm ae, y-k=0$. Hence foci are $(h \pm ae, k)$

(vii) Equation of directrices of the hyperbola are $X = \pm\frac{a}{e}$, i.e., $x - h = \pm\frac{a}{e}$

Hence directrices are $x = h \pm \frac{a}{e}$

(viii) Length of latus rectum $= \frac{2b^2}{a}$

(ix) Coordinate of ends of latus rectum are given by

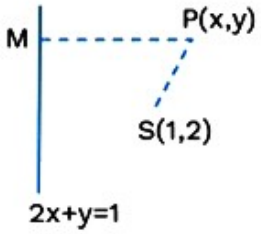
$X = ae, Y = \pm\frac{b^2}{a}$ i.e. $x - h = \pm ae, y - k = \pm\frac{b^2}{a}$

$\therefore$ ends of LR is $\left(h \pm ae, k \pm \frac{b^2}{a}\right)$

Illustrations:

Q. **Find the equation of the hyperbola whose directrix is $2x + y = 1$, focus $(1,2)$ and eccentricity $\sqrt{3}$.**

Sol. Let P(x, y) be any point on the hyperbola.
Draw PM perpendicular from P on the directrix.
Then by definition SP = e.PM



$\Rightarrow (SP)^2 = e^2(PM)^2$

$$\Rightarrow (x-1)^2 + (y-2)^2 = 3\left\{\frac{2x+y-1}{\sqrt{4+1}}\right\}^2$$

$$\Rightarrow 5\left(x^2 + y^2 - 2x - 4y + 5\right) = 3\left(4x^2 + y^2 + 1 + 4xy - 2y - 4x\right)$$

$$\Rightarrow 7x^2 - 2y^2 + 12xy - 2x + 14y - 22 = 0$$

Which is the required hyperbola.

Q. **Find the lengths of transverse axis, conjugate axis, eccentricity, the co-ordinates of foci, vertices, lengths of the latus-rectum and equations of the directrices of the following hyperbolas.**

(i) $9x^2 - y^2 = 1$ **(ii)** $16x^2 - 9y^2 = -144$

(iii) $7y^2 - 9x^2 + 54x - 28y - 116 = 0$ **(iv)** $9x^2 - 16y^2 - 72x + 96y - 144 = 0$

Sol. **(i)** The equation $9x^2 - y^2 = 1$ can be written as $\frac{x^2}{(1/9)} - \frac{y^2}{1} = 1 \Rightarrow a = \frac{1}{3}, b = 1$

The length of transverse, axis $= 2a = \frac{2}{3}$

The length of conjugate axis $= 2b = 2$.

Eccentricity $e = \sqrt{\left(1 + \frac{b^2}{a^2}\right)} = \sqrt{1 + \frac{1}{(1/9)}} = \sqrt{10}$

The co-ordinates of the foci are $(\pm ae, 0)$ i.e., $\left(\pm \frac{\sqrt{10}}{3}, 0\right)$

The length of latus - rectum $= \frac{2b^2}{a} = \frac{2(1)^2}{1/3} = 6$

The equations of the directrices are $x = \pm \frac{a}{e}$ i.e., $x = \pm \frac{1/3}{\sqrt{10}}$ or $x = \pm \frac{1}{3\sqrt{10}}$

(ii) The equation $16x^2 - 9y^2 = -144$ can be written as $\frac{x^2}{9} - \frac{y^2}{16} = -1$.

a=3, b=4, This conjugate hyperbola
The length of transverse axis =2b=8.
The length of conjugate axis =2a=6

Eccentricity $e = \sqrt{\left(1 + \frac{a^2}{b^2}\right)} = \sqrt{\left(1 + \frac{9}{16}\right)} = \frac{5}{4}$

The co-ordinates of the foci are $(0, \pm be)$ i.e., $(0, \pm 5)$

The co-ordinates of the vertices are $(0, \pm b)$ i.e., $(0, \pm 4)$

The length of latus-rectum $= \frac{2a^2}{b} = \frac{2(3)^2}{4} = \frac{9}{2}$

The equation of directrices are $y = \pm \frac{b}{e} = \pm \frac{4}{(5/4)} \Rightarrow y = \pm \frac{16}{5}$.

(iii) We have $7y^2 - 9x^2 + 54x - 28y - 116 = 0$

or $7\left(y^2 - 4y\right) - 9\left(x^2 - 6x\right) - 116 = 0$

or $7\left(y^2 - 4y + 4\right) - 9\left(x^2 - 6x + 9\right) = 116 + 28 - 81$

or $7(y-2)^2 - 9(x-3)^2 = 63$

or $\frac{(y-2)^2}{9} - \frac{(x-3)^2}{7} = 1$

or $\frac{Y^2}{9} - \frac{X^2}{7} = 1$

where $X = x-3$ and $Y = y-2$.

This equation represents conjugate hyperbola. Comparing it with $\frac{Y^2}{b^2} - \frac{X^2}{a^2} = 1$

We get $b^2 = 9$ and $a^2 = 7$

$\therefore b = 3$ and $a = \sqrt{7}$

Centre: $X = 0, Y = 0$ i.e. $x - 3 = 0, y - 2 = 0$

$\therefore$ Centre is (3,2)

Length of transverse axis $= 2b = 6$

Length of conjugate axis $= 2a = 2\sqrt{7}$

The eccentricity e is given by

$$e = \sqrt{1 + \frac{a^2}{b^2}} = \sqrt{1 + \frac{7}{9}} = \frac{4}{3}.$$

The length of latus-rectum $= \frac{2a^2}{b} = \frac{2(7)}{3} = \frac{14}{3}$

The co-ordinates of foci $X = 0, Y = \pm$ be

$\Rightarrow \quad x - 3 = 0, y - 2 = \pm 3 \times \frac{4}{3}$ or $(3, 2 \pm 4)$

i.e. (3, −2) and (3, 6)

The co-ordinates of vertices are

Or $\quad X = 0, Y = \pm b$

Or $\quad (3, 2 \pm 3)$

Or $\quad$ vertices are (3, −1) and (3, 5)

The equation of directrices are $Y = \pm \frac{b}{e}$

$$\Rightarrow \quad y - 2 = \pm \frac{3}{4/3} \Rightarrow y = \left(2 \pm \frac{9}{4}\right)$$

i.e. $y = \frac{17}{4}$ and $y = \frac{-1}{4}$

(iv) Equation can be rewritten as $\frac{(x-4)^2}{4^2} - \frac{(y-3)^2}{3^2} = 1$

So a = 4, b = 3

$b^2 = a^2\left(e^2 - 1\right)$ gives $e = \frac{5}{4}$.

Foci: x−4 = ±ae,
y−3=0
foci are (9, 3), (−1, 3)

Centre: x − 4 = 0, y - 3 = 0,
i.e. centre is (4, 3)

Directrices : $x - 4 = \pm\frac{a}{e}$, i.e. $x - 4 = \pm\frac{16}{5}$

Hence, directrices are 5x−36=0 ; 5 x−4=0.

Latus rectum: $\frac{2b^2}{a} = 2 \times \frac{9}{4} = \frac{9}{2}$

Q. **Find the equation of hyperbola**
(a) Whose axes are coordinate axes and the distances of one of its vertices from the foci are 3 and 1;
(b) whose centre is (1,0), focus is (6,0) and transverse axis is 6;
(c) whose centre is (3,2), one focus is (5,2) and one vertex is (4,2);
(d) whose centre is (−3,2), one vertex is (−3,4) and eccentricity is $\frac{5}{2}$
(e) whose foci are (4,2) and (8,2) and eccentricity is 2.

Sol. **(a)** ae − a = 1 and ae + a = 3

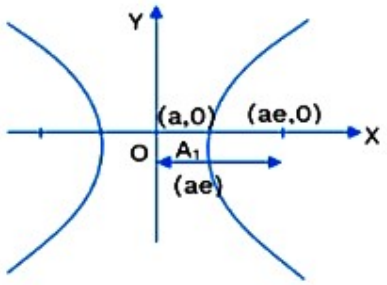
$\therefore \quad \frac{e+1}{e-1} = 3$

$\Rightarrow$ e = 2 and a = 1

Also from $b^2 = a^2\left(e^2 - 1\right) = 3$,

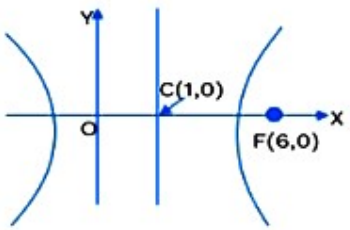
the equation is $x^2 - \frac{y^2}{3} = 1$

or $\frac{x^2}{3} - \frac{y^2}{1} = -1$



(b) Equation of hyperbola with centre (1,0) is

$\frac{(x-1)^2}{a^2} - \frac{y^2}{a^2\left(e^2 - 1\right)} = 1$



Given a=3 and ae=5 ; hence $e = \frac{5}{3}$.

Therefore, Eq. (i) becomes

$$\frac{(x-1)^2}{9} - \frac{y^2}{9\left(\frac{25}{9}-1\right)} = 1$$

$$\Rightarrow \quad \frac{(x-1)^2}{9} - \frac{y^2}{16} = 1$$

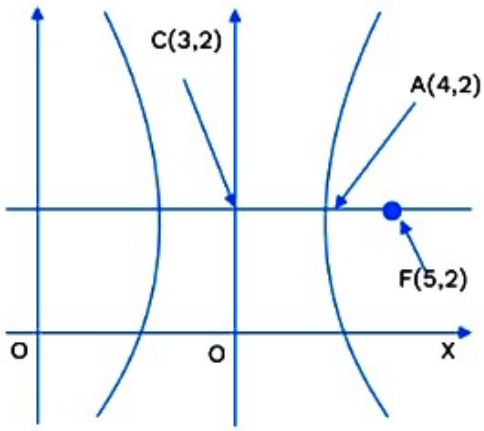
(c) Equation of hyperbola with centre (3,2) is

$$\frac{(x-3)^2}{a^2} - \frac{(y-2)^2}{a^2\left(e^2-1\right)} = 1$$

with axis parallel to x-axis
a = distance between centre and vertex =1
ae = distance between centre and focus.
Hence, ae = 2 $\Rightarrow$ e = 2
Hence, the equation is

$$\frac{(x-3)^2}{1} - \frac{(y-2)^2}{(4-1)} = 1$$

$$\Rightarrow \quad (x-3)^2 - \frac{(y-2)^2}{3} = 1$$



(d) Equation of the hyperbola is

$$\frac{(x+3)^2}{a^2} - \frac{(y-2)^2}{b^2} = -1$$

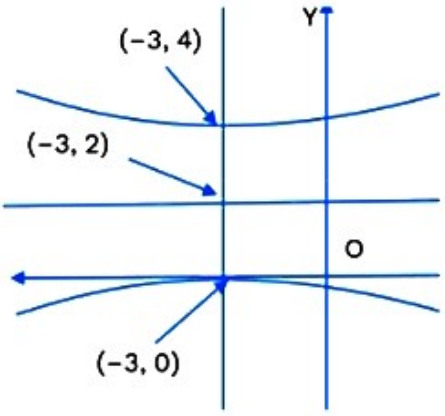
(as the line joining centre to the vertex is parallel to y-axis)
Now b=2 (distance between centre and vertex)

$$\text{Focus} \quad = be = 2 \times \frac{5}{2} = 5$$

$$\text{Also} \quad = b^2\left(e^2-1\right) \; = 4\left(\frac{25}{4}-1\right) = 21$$

Therefore, the required equation is

$$\frac{(x+3)^2}{21} - \frac{(y-2)^2}{4} = -1$$



(e) Line joining the foci is parallel to x-axis.
Distance between the two foci =4=2ae
Hence, a = 1 as e = 2

$b^2 = a^2(e^2-1) = 3$

Now centre is midpoint of the line joining the foci which is (6,2)

Hence, equation of the hyperbola is

$$\frac{(x-6)^2}{1} - \frac{(y-2)^2}{3} = 1$$

Q. **The equation $16x^2 - 3y^2 - 32x + 12y - 44 = 0$ represents a hyperbola**

(A) the length of whose transverse axis is $2\sqrt{3}$

(B) the length of whose conjugate axis is 8

(C) whose centre is (1,2)

(D) whose eccentricity is $\frac{\sqrt{19}}{3}$

Sol. **(ABC)**

We have, $16\left(x^2-2x\right) - 3\left(y^2-4y\right) = 44 \Rightarrow 16(x-1)^2 - 3(y-2)^2 = 48$

$$\Rightarrow \frac{(x-1)^2}{3} - \frac{(y-2)^2}{16} = 1$$

Centre is $x-1=0$ & $y-2=0 \Rightarrow (1,2)$

$a^2 = 3 \Rightarrow a = \sqrt{3}, b^2 = 16 \Rightarrow b = 4$

$\therefore$ $l(TA) = 2a = 2\sqrt{3}$ and $l(CA) = 2b = 2.4 = 8$.

This equation represents a hyperbola with eccentricity given

$$e = \sqrt{1+\left(\frac{\text{Conjugate axis}}{\text{Transverse axis}}\right)^2} = \sqrt{1+\left(\frac{4}{\sqrt{3}}\right)^2} = \sqrt{\frac{19}{3}}$$

Q. **The equation $9x^2 - 16y^2 - 18x + 32y - 151 = 0$ represent a hyperbola**

(A) The length of the transverse axes is 4

(B) Length of latus rectum is 9

(C) Equation of directrix is $x = \frac{21}{5}$ and $x = -\frac{11}{5}$

(D) None of these

Sol. **(C)**

We have $9x^2 - 16y^2 - 18x + 32y - 151 = 0$

$9\left(x^2-2x\right) - 16\left(y^2-2y\right) = 151$

$9\left(x^2-2x+1\right) - 16\left(y^2-2y+1\right) = 144$

$9(x-1)^2 - 16(y-1)^2 = 144$

$\frac{(x-1)^2}{16} - \frac{(y-1)^2}{9} = 1$

Shifting the origin at (1,1) without rotating the axes $\frac{X^2}{16} - \frac{Y^2}{9} = 1$

Where $X = x - 1$ and $Y = y - 1$

This is of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Where $a^2 = 16$ and $b^2 = 9$ so
The length of the transverse axes $=2a = 8$; $l(CA) = 2b = 6$.

The length of the latus rectum $= \frac{2b^2}{a} = \frac{2 \cdot 9}{4} = \frac{9}{2}$ and $e = \sqrt{1 + \frac{b^2}{a^2}} = \sqrt{1 + \frac{9}{16}} = \frac{5}{4}$

The equation of the directrix $X = \pm \frac{a}{e}$

$x - 1 = \pm \frac{16}{5} \Rightarrow x = \pm \frac{16}{5} + 1$

$x = \frac{21}{5}$ and $x = -\frac{11}{5}$

Equation of directrix is $x = \frac{21}{5}$ and $x = -\frac{11}{5}$

Q. **If the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ and hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ coincide, then the value b^2 is**

(A) 1 **(B) 5** **(C) 7** **(D)9**

Sol. **(C)**

For hyperbola $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25} \Rightarrow \frac{x^2}{144/25} - \frac{y^2}{81/25} = 1$

$e^2 = 1 + \frac{b^2}{a^2} = 1 + \frac{81}{144} = \frac{225}{144}; e = \frac{15}{12} = \frac{5}{4}$

Hence the foci are

$(\pm ae, 0) = \left(\pm \frac{12}{5} \cdot \frac{5}{4}, 0 \right) = (\pm 3, 0)$

Now the foci coincide therefore for ellipse.

$ae = 3$ or $a^2e^2 = 9$ or $a^2\left(1-\frac{b^2}{a^2}\right) = 9$

$a^2 - b^2 = 9$ or $16 - b^2 = 9 \Rightarrow b^2 = 7$

8. **The foci of a hyperbola coincides with the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ Find the equation of the hyperbola if its eccentricity is 2.**

Sol. The given ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ or $\frac{x^2}{5^2}+\frac{y^2}{3^2}=1$ ...(1)

Here $5 > 3$ $\therefore$ a=5 and b=3

$\therefore$ The foci of the ellipse are on the x -axis.

$\therefore$ Eccentricity of the ellipse, $e = \sqrt{1-\frac{b^2}{a^2}} = \sqrt{1-\frac{9}{25}} = \frac{4}{5}$

$\therefore$ Foci of the ellipse are $(\pm ae, 0)$ or $(\pm 4, 0)$

Let the equation of the hyperbola be $\frac{x^2}{A^2}-\frac{y^2}{B^2}=1$

$\therefore$ Foci are $(\pm Ae, 0)$ or $(\pm 2A, 0)$ $[\because e = 2]$

$\therefore$ 2 A=4 i.e., A=2 and $B = A\sqrt{e^2-1} = 2\sqrt{(2)^2-1} = 2\sqrt{3}$

The equation of the hyperbola is $\frac{x^2}{(2)^2}-\frac{y^2}{(2\sqrt{3})^2}=1$ or $\frac{x^2}{4}-\frac{y^2}{12}=1$

Or $3x^2 - y^2 = 12$

Q. **Find the eccentricity of the hyperbola whose latus-rectum is half of its transverse axis.**

Sol. Let the equation of hyperbola be $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.

Then transverse axis = 2a and latus-rectum $= \frac{2b^2}{a}$

According to question $\frac{2b^2}{a} = \frac{1}{2}(2a)$

$\Rightarrow 2b^2 = a^2$ $\left(\because b^2 = a^2\left(e^2-1\right)\right)$

$\Rightarrow 2a^2\left(e^2-1\right)=a^2 \quad \Rightarrow \quad 2e^2-2=1 \quad \Rightarrow \quad e^2=\frac{3}{2}$

$\therefore e=\sqrt{\frac{3}{2}}$

Hence the required eccentricity is $\sqrt{\frac{3}{2}}$.

Q. **If $f(x)=x^3+\alpha x^2+\beta x+\gamma$, where α, β, γ are rational numbers and two roots of $f(x)=0$ are eccentricities of a parabola and a rectangular hyperbola, then $\alpha+\beta+\gamma$ is equal to**

(A) –1 **(B) 0** **(C) 1** **(D) 2**

Sol. **(A)**

Two confirmed roots are 1 and $\sqrt{2}$.

So another has to be $-\sqrt{2}$

$(x-\sqrt{2})(x-1)(x+\sqrt{2})=\left(x^2-2\right)(x-1)$

$=x^3-x^2-2x+2$

$\alpha+\beta+\gamma=-1-2+2=-1$

Q. **(a) Find the equation of the hyperbola whose foci are $(0,\pm\sqrt{10})$ and which passes through the point (2,3).**

(b) Find the equation of the hyperbola, the length of whose latus rectum is 8, eccentricity is $\frac{3}{\sqrt{5}}$ and whose transverse and conjugate axes are along the x and y axes respectively.

(c) Find the equation of the hyperbola whose eccentricity is $\sqrt{2}$ and the distance between the foci is 16, taking transverse and conjugate axes of the hyperbola as x and y-axes respectively.

Sol. **(a)** The foci of the hyperbola are at $(0,\pm\sqrt{10})$. These are on the y-axis Clearly, centre of the hyperbola will be (0,0)

Let the equation of the hyperbola $\frac{y^2}{b^2}-\frac{x^2}{a^2}=1$ ...(1)

Where $a,b>0$ and $a^2=b^2\left(e^2-1\right) \Rightarrow a^2+b^2=a^2e^2$ i.e. $a^2+b^2=10$

$$\frac{9}{b^2}-\frac{4}{10-b^2}=1 \Rightarrow b^4-23b^2+90=0 \Rightarrow b^2=18,\ 5$$

when $b^2=18$ then $a^2=-8$ not possible ; when $b^2=5 \Rightarrow a^2=5$ possible.

Hence the equation of hyperbola is $\frac{y^2}{5}-\frac{x^2}{5}=1$ or $y^2-x^2=5$

(b) Given, $8=\frac{2b^2}{a}=\frac{2a^2\left(e^2-1\right)}{a}$

$$\Rightarrow \quad 8=2a\left(e^2-1\right)=2a\left[\left(\frac{3}{\sqrt{5}}\right)^2-1\right]=2a\left[\frac{9}{5}-1\right]=\frac{8a}{5}$$

$$\Rightarrow \quad a=5$$

Again $8=\frac{2b^2}{a}=8\times 5 \quad \therefore b^2=20$

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1 \quad \text{or} \quad \frac{x^2}{25}-\frac{y^2}{20}=1$$

(c) Let the equation of the hyperbola be

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1 \qquad \ldots(1)$$

The coordinates of the foci are (ae, 0) and (−ae, 0)

Given $2ae=16$ or $2\cdot a\sqrt{2}=16 \quad \therefore a=4\sqrt{2}$

Also $b^2=a^2\left(e^2-1\right)=32(2-1)=32 \quad [\because e=\sqrt{2}]$

The required equation of the hyperbola is $\frac{x^2}{32}-\frac{y^2}{32}=1$ or $x^2-y^2=32$

Q. **Prove that the point $\left\{\frac{a}{2}\left(t+\frac{1}{t}\right),\frac{b}{2}\left(t-\frac{1}{t}\right)\right\}$ lies on the hyperbola for all values of t($t\neq 0$).**

Sol. Let $x=\frac{a}{2}\left(t+\frac{1}{t}\right)$ or $\frac{2x}{a}=t+\frac{1}{t}$ or $\left(\frac{2x}{a}\right)^2=t^2+\frac{1}{t^2}+2$ ...(1)

And $y=\frac{b}{2}\left(t-\frac{1}{t}\right)$ or $\frac{2y}{b}=t-\frac{1}{t}$ or $\left(\frac{2y}{b}\right)^2=t^2+\frac{1}{t^2}-2$ ...(2)

Subtracting (2) from (1), $\frac{4x^2}{a^2}-\frac{4y^2}{b^2}=4$ or $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$Which is hyperbola.

Q. **If e and e′ are the eccentricities of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and its conjugate hyperbola, prove that $\frac{1}{e^2}+\frac{1}{e'^2}=1$.**

Sol. Given hyperbola is $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ ...(1)

The eccentricity e of hyperbola (1) is given by $b^2=a^2\left(e^2-1\right)$

$\therefore \quad \frac{b^2}{a^2}=e^2-1$ ...(2)

The equation of the conjugate hyperbola is $\frac{y^2}{b^2}-\frac{x^2}{a^2}=1$

$\therefore$ Its eccentricity e′ is given by $a^2=b^2\left(e'^2-1\right)$

$\therefore \quad \left(e'^2-1\right)=\frac{a^2}{b^2}$ ...(3)

Multiple (2) and (3), we get

$\left(e^2-1\right)\times\left((e')^2-1\right)=\frac{a^2}{b^2}\times\frac{b^2}{a^2}$

$\therefore e^2e'^2-e^2-e'^2+1=1$

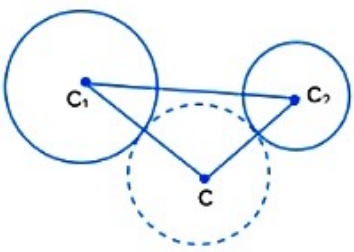
$\therefore \quad e^2e'^2=e^2+e'^2$

$\therefore \quad 1=\frac{1}{e^2}+\frac{1}{e'^2}$

Q. **Two circles are given such that they neither intersect nor touch. Then identify the locus of centre of variable circle which touches both the circles externally.**

sol. In the figure circles with solid line have centres C_1 and C_2 and radii r_1 and r_2.
Let the circle with dotted line is variable circle which touches given two circles as explained in the question which has centre C and radius r.
Now, $CC_2=r+r_2$
and $CC_1=r_1+r$
Hence $CC_1-CC_2=r_1-r_2$ (=constant)

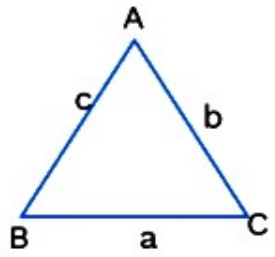
Then locus of C is hyperbola whose foci are C_1 and C_2



Q. **If base of triangle and ratio of tangents of half of base angles are given, then identify the locus of opposite vertex.**

Sol. In ΔABC, base BC = a is given

Also $\dfrac{\tan\frac{B}{2}}{\tan\frac{C}{2}} = k$(constant)



$$\Rightarrow \frac{\sqrt{\frac{s(s-b)}{(s-a)(s-c)}}}{\sqrt{\frac{s(s-c)}{(s-a)(s-b)}}} = k$$

$$\Rightarrow \frac{s-b}{s-c} = k$$

$$\Rightarrow \frac{(s-b)-(s-c)}{(s-b)+(s-c)} = \frac{k-1}{k+1}$$

$$\Rightarrow \frac{c-b}{2s-(b+c)} = \frac{k-1}{k+1}$$

$$\Rightarrow c-b = a\frac{k-1}{k+1} = \text{ constant}$$

$\therefore$ AB – AC= constant

So, locus of A is a hyperbola.

Q. **If two points P and Q on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ whose centre is C, are such that CP is perpendicular to CQ, a < b, then prove that $\frac{1}{(CP)^2}+\frac{1}{(CQ)^2}=\frac{1}{a^2}-\frac{1}{b^2}$**

Sol. Let P(x, y) be any point on the given hyperbola. Then equation of CP is $y = mx$

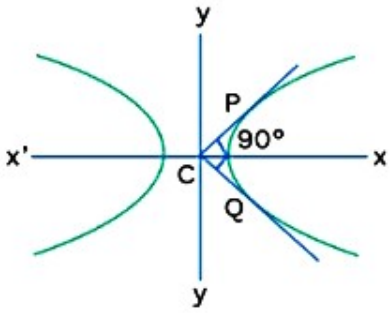
Solving $y = mx$ and

$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$$

$$\frac{x^2}{a^2}-\frac{m^2x^2}{b^2}=1$$

$$x^2 = \frac{a^2b^2}{b^2-a^2m^2}$$

and $y^2 = m^2x^2 = \dfrac{a^2b^2m^2}{b^2-a^2m^2}$



$$x^2 + y^2 = \frac{a^2b^2}{(b^2 - a^2m^2)}(1+m^2)$$

$$(CP)^2 = x^2 + y^2 = \frac{a^2b^2(1+m^2)}{(b^2 - a^2m^2)}$$

$$\frac{1}{(CP)^2} = \frac{b^2 - a^2m^2}{a^2b^2(1+m^2)} \qquad ...(i)$$

and equation of CQ is $y = -x/m$

replacing m by $-\frac{1}{m}$ in Eq. (i)

$$\text{then } \frac{1}{(CQ)^2} = \frac{b^2 - a^2\left(-\frac{1}{m}\right)^2}{a^2b^2\left(1+\left(-\frac{1}{m}\right)^2\right)} = \frac{b^2m^2 - a^2}{a^2b^2(1+m^2)} \qquad ...(ii)$$

Adding equation (i) and (ii), then

$$\frac{1}{(CP)^2} + \frac{1}{(CQ)^2} = \frac{b^2(1+m^2) - a^2(1+m^2)}{a^2b^2(1+m^2)} = \frac{b^2 - a^2}{a^2b^2} = \frac{1}{a^2} - \frac{1}{b^2}$$

Q. **For the hyperbola whose latus rectum subtends a right angle at its centre find the eccentricity.**

Sol. $m_{CP} \times m_{CQ} = -1$

$$\Rightarrow \left(\frac{\frac{b^2}{a}}{ae}\right) \times \left(\frac{\frac{-b^2}{a}}{ae}\right) = -1$$

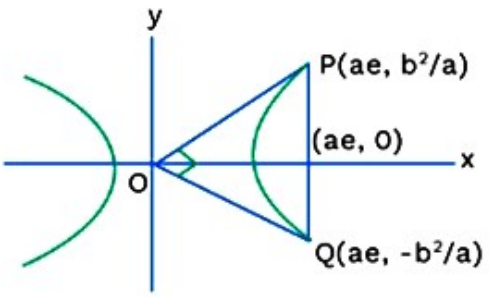
$$\Rightarrow \frac{b^4}{a^4e^2} = 1$$

$$\Rightarrow (e^2 - 1)^2 = e^2$$

$$\Rightarrow e^4 - 3e^2 + 1 = 0$$

$$\Rightarrow e^2 = \frac{3 \pm \sqrt{5}}{2}$$

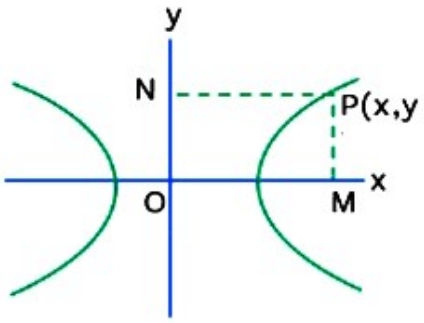
$$\Rightarrow e^2 = \frac{3+\sqrt{5}}{2} \Rightarrow e = \frac{\sqrt{5}+1}{2}$$



Equation of a Hyperbola Referred to Two Perpendicular Lines

Consider the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ as shown in figure. Let P(x, y) be any point on the hyperbola. Then PM = y and PN = x.

$\therefore \frac{PN^2}{a^2}-\frac{PM^2}{b^2}=1$

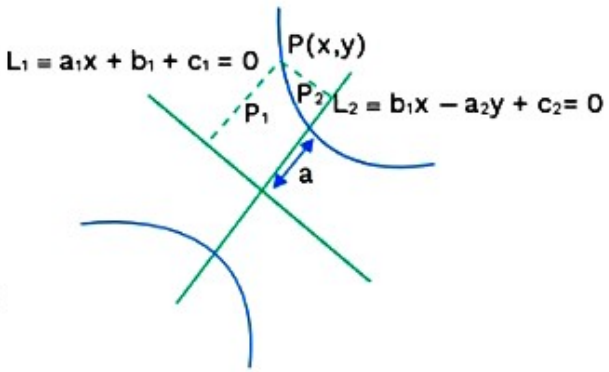


It follows from this that if perpendicular distance p_1 and p_2 of a moving point P(x, y) from two mutually perpendicular coplanar straight lines $L_1 \equiv a_1x + b_1y + c_1 = 0$, $L_2 \equiv b_1x - a_1y + c_2 = 0$, respectively, satisfy the equation

$\frac{p_1^2}{a^2}-\frac{p_2^2}{b^2}=1$

$$\frac{\left(\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}}\right)^2}{a^2}-\frac{\left(\frac{b_1x-a_1y+c_2}{\sqrt{b_1^2+a_1^2}}\right)^2}{b^2}=1$$

then the locus of point P denotes an hyperbola in the plane of the given lines such that



(i) The centre of the hyperbola is the point of intersection of lines $L_1 = 0$ and $L_2 = 0$;

(ii) The transverse axis lies along $L_2 = 0$ and the conjugate axis lies along $L_1 = 0$

(iii) The length of the transverse and conjugate axes arc 2a and 2b.

Q. **Find the coordinates of the foci and the centre of the hyperbola**

$$\frac{(3x-4y-12)^2}{100}-\frac{(4x+3y-12)^2}{225}=1.$$

Sol. Let $3x - 4y - 12 = X$ and $4x + 3y - 12 = Y$.

$\frac{X^2}{10^2}-\frac{Y^2}{15^2}=1$

(note that X and Y are two perpendicular lines)

Centre is the point of intersection of X = 0 and Y = 0

Hence, $3x - 4y = 12$ ... (i)

$4x + 3y = 12$ ... (ii)

Solving (i) and (ii), we get $x = \frac{84}{25}, y = -\frac{12}{25}$

Now, $e^2 = 1 + \frac{225}{100} = \frac{325}{100}$

$$e = \frac{\sqrt{13}}{2}$$

Also, $a = 10, b = 15$

Focus is $(ae, 0)$

Hence, $X = ae$

and $Y = 0$

$\Rightarrow 3x - 4y - 12 = ae$

and $4x + 3y - 12 = 0$

Solving, we get

$$x = \frac{84 + 3ae}{25} = \frac{84 + 15\sqrt{13}}{25}$$

$$y = \frac{-12 + 15\sqrt{13}}{25}$$

Hence, focus is $\left(\frac{84 + 15\sqrt{3}}{25}, \frac{-12 + 15\sqrt{13}}{25}\right)$

Q. **The equations of the transverse and conjugate axes of a hyperbola are respectively $3x + 4y - 7 = 0$, $4x - 3y + 8 = 0$ and their respective lengths are 4 and 6. Find the equation of the hyperbola.**

Sol. The equation of the required hyperbola is

$$\frac{\left(\frac{3x + 4y - 7}{\sqrt{(3^2 + 4^2)}}\right)^2}{\left(\frac{4}{2}\right)^2} - \frac{\left(\frac{4x - 3y + 8}{\sqrt{4^2 + (-3)^2}}\right)^2}{\left(\frac{6}{2}\right)^2} = 1$$

$$\frac{1}{100}(3x + 4y - 7)^2 - \frac{1}{225}(4x - 3y + 8)^2 = 1$$

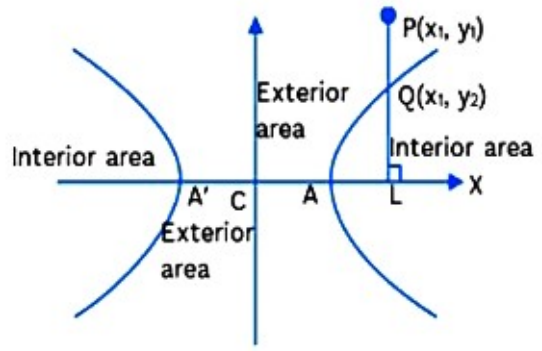
$\Rightarrow 9(3x + 4y - 7)^2 - 4(4x - 3y + 8)^2 = 900$

$\Rightarrow 9(9x^2 + 16y^2 + 49 + 24xy - 42x - 56y) - 4(16x^2 + 9y2 + 64 - 24xy + 64x - 48y) = 900$

$\Rightarrow 17x^2 + 312xy + 108y^2 \quad 634x - 312y - 715 = 0$

POSITION OF A POINT WITH RESPECT TO HYPERBOLA

Let $S(x,y) \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1$, be the given hyperbola and $P(x_1, y_1)$ is the given point.



Point to Remember!!!

(i) If $S(x_1, y_1) > 0$ then $P(x_1, y_1)$ lie inside the hyperbola

(ii) If $S(x_1, y_1) < 0$ then $P(x_1, y_1)$ lie outside the hyperbola

(iii) If $S(x_1, y_1) = 0$ then $P(x_1, y_1)$ lie on the hyperbola

Illustration:

Q. **Find the position of the point (5,–4) relative to the hyperbola $9x^2 - y^2 = 1$**

Sol. Here $S(x,y) \equiv 9x^2 - y^2 - 1$

and $S(5,-4) = 9(5)^2 - (-4)^2 - 1 = 255 - 16 - 1 = 208 > 0$

So the point (5,–4) inside the hyperbola $9x^2 - y^2 = 1$.

INTERSECTION OF A LINE AND A HYPERBOLA

Let the hyperbola be $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ ...(1)

and the given line be $y = mx + c$ ...(2)

The point of intersection of line and hyperbola can be obtained by solving (1) and (2), therefore

Eliminating from (1) and (2), we get $\frac{x^2}{a^2} - \frac{(mx+c)^2}{b^2} = 1$

$\Rightarrow b^2x^2 - a^2(mx+c)^2 = a^2b^2$

$\Rightarrow (a^2m^2 - b^2)x^2 + 2mca^2x + a^2(b^2 + c^2) = 0$ ...(3)

Above equation is a quadratic in x and gives two values of x, It shows that every straight line will cut the hyperbola in two points, may be real, coincident or imaginary according as discriminant of (3) $>, =, < 0$

i.e. $4m^2c^2a^4 - 4\left(a^2m^2 - b^2\right)a^2\left(b^2 + c^2\right) >, =, < 0$

or $-a^2m^2 + b^2 + c^2 >, =, < 0$

or $c^2 >, =, < a^2m^2 - b^2$ ...(4)

EQUATIONS OF TANGENT

(1) Point form:

The equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at (x_1, y_1) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$

Since point (x_1, y_1) lie on the hyperbola therefore we can use standard substitution to obtain the equation of tangent.

(2) Parametric form:

The equation of tangent to the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at $(a\sec\theta, b\tan\theta)$ is $\frac{x}{a}\sec\theta - \frac{y}{b}\tan\theta = 1$

Note: Point of intersection of the tangents at θ_1 & θ_2 is $x = a\frac{\cos\frac{\theta_1 - \theta_2}{2}}{\cos\frac{\theta_1 + \theta_2}{2}}, y = b\frac{\sin\frac{\theta_1 + \theta_2}{2}}{\cos\frac{\theta_1 + \theta_2}{2}}$

(3) Slope form:

Given hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ ...(1)

and the given line $y = mx + c$ ... (2)

If the given line touches hyperbola then solve (1) and (2), we get

$$\frac{x^2}{a^2} - \frac{(mx + c)^2}{b^2} = 1$$

or $b^2x^2 - a^2(mx + c)^2 = a^2b^2$

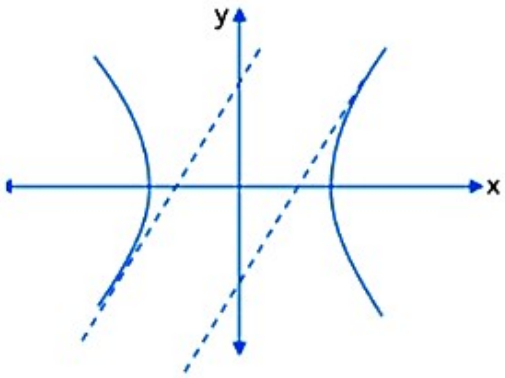
Or $\left(b^2 - a^2m^2\right)x^2 - 2a^2mcx - a^2\left(c^2 + b^2\right) = 0$ ...(3)

Since line (2) will be tangent to hyperbola (1) if roots of equation (3) are equal i.e. D = 0

Point to Remember!!!

(i) Equation of any tangent to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ may be taken as $y = mx + \sqrt{a^2m^2 - b^2}$ or $y = mx - \sqrt{a^2m^2 - b^2}$ The co-ordinates of the points of contact are $\left(\pm\frac{a^2m}{\sqrt{\left(a^2m^2 - b^2\right)}}, \pm\frac{b^2}{\sqrt{\left(a^2m^2 - b^2\right)}}\right)$

(ii) The equation of any tangent to the hyperbola $\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$ may be taken as $(y - k) = m(x - h) \pm \sqrt{a^2m^2 - b^2}$



$$4a^4m^2c^2 + 4a^2\left(b^2 - a^2m^2\right)\left(c^2 + b^2\right) = 0$$

or $a^2m^2c^2 + b^2c^2 - a^2c^2m^2 + b^4 - a^2b^2m^2 = 0$

or $b^2c^2 + b^4 - a^2b^2m^2 = 0$

or $c^2 + b^2 - a^2m^2 = 0$

or $c^2 = a^2m^2 - b^2$ or $c = \pm\sqrt{a^2m^2 - b^2}$.

This is the required condition of tangency

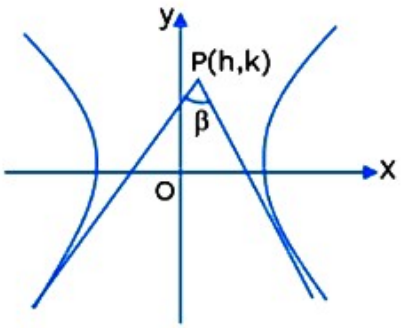
Tangents drawn from outside point :

$y = mx \pm \sqrt{a^2m^2 - b^2}$ is a tangent to the standard hyperbola.

If above tangent passes through (h, k) then

$(k - mh)^2 = a^2m^2 - b^2$

$\left(h^2 - a^2\right)m^2 - 2kmh + k^2 + b^2 = 0$



Above equation is quadratic in m therefore it has two roots m_1 and m_2.

Hence passing through a given point (h, k) there is a maximum of two tangents can be drawn to the hyperbola

Therefore $m_1 + m_2 = \dfrac{2kh}{h^2 - a^2}$ ...(A)

$$m_1m_2 = \frac{k^2 + b^2}{h^2 - a^2} \quad \text{...(B)}$$

Equations (A) and (B) are used to find the locus of the point of intersection of a pair of tangents which enclose an angle β.

Now $\tan^2\beta = \dfrac{\left(m_1 + m_2\right)^2 - 4m_1m_2}{\left(1 + m_1m_2\right)^2}$

(substituting the values of $m_1 + m_2$ and m_1m_2 to get the locus)

If $\beta = 90°$ then $m_1m_2 = -1$ hence from (B)

$k^2 + b^2 = a^2 - h^2$

$x^2 + y^2 = a^2 - b^2$

which is the equation of director circle, of the given hyperbola.

Illustrations :

Q. **Find the equation of the tangent to the hyperbola $x^2 - 4y^2 = 36$ which is perpendicular to the line $x - y + 4 = 0$**

Sol. Let be the slope of the tangent. Since the tangent is perpendicular to the line $x - y + 4 = 0$

$m \times 1 = -1 \Rightarrow m = -1$

Given hyperbola $x^2 - 4y^2 = 36$ or $\frac{x^2}{36} - \frac{y^2}{9} = 1, a^2 = 36, b^2 = 9$

$\therefore$ tangent to above hyperbola is $y = mx \pm \sqrt{36m^2 - 9}$

Here $m = -1$

equation of tangent is $y = -x \pm \sqrt{27}$ or $x + y \pm 3\sqrt{3} = 0$

Q. **For what value of λ does the line $y = 2x + \lambda$ touches the hyperbola $16x^2 - 9y^2 = 144$?**

Sol. Equation of hyperbola is $16x^2 - 9y^2 = 144$ or $\frac{x^2}{9} - \frac{y^2}{16} = 1$

Comparing this with $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ we get $a^2 = 9, b^2 = 16$

and comparing this line $y = 2x + \lambda$ with $y = mx + c$.

$\therefore m = 2, c = \lambda$

If the line $y=2x+\lambda$ touches the hyperbola $16x^2 - 9y^2 = 144$ then $c^2 = a^2m^2 - b^2$

$\Rightarrow \lambda^2 = 9(2)^2 - 16 = 36 - 16 = 20$

$\therefore \lambda = \pm 2\sqrt{5}$

Q. **Find the area of a triangle formed by the lines $x-y=0$, $x+y=0$ and any tangent to the hyperbola $x^2 - y^2 = a^2$**

Sol. Any tangent to the hyperbola at $P(a\sec\theta, a\tan\theta)$ is

$x\sec\theta - y\tan\theta = a$ ...(i)

Also, $x - y = 0$ ...(ii)

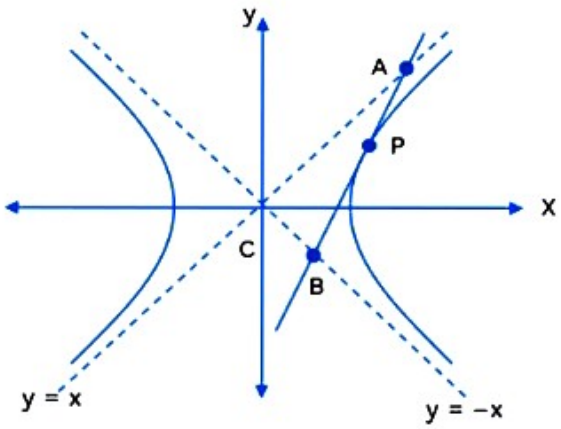
$$x + y = 0 \qquad \text{...(iii)}$$

Solving the above three lines in pairs, we get the point A,B,C as

$$\left(\frac{a}{\sec\theta - \tan\theta}, \frac{a}{\sec\theta - \tan\theta}\right), \left(\frac{a}{\sec\theta + \tan\theta}, \frac{-a}{\sec\theta + \tan\theta}\right) \text{ and } (0,0)$$

Since the one vertex is the origin

therefore the area of the triangle ABC is



$$= \frac{1}{2}\left|(x_1y_2 - x_2y_1)\right|$$

$$= \left|\frac{a^2}{2}\left(\frac{-1}{\sec^2\theta - \tan^2\theta} - \frac{1}{\sec^2\theta - \tan^2\theta}\right)\right|$$

$$= \left|\frac{a^2}{2}(-2)\right| = a^2$$

Q. **Find the equation and the length of the common tangents to hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $\frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$**

Sol. Tangent at (a secϕ, b tan ϕ) on the 1st hyperbola is

$$\frac{x}{a}\sec\phi - \frac{y}{b}\tan\phi = 1 \qquad \text{...(1)}$$

Similarly tangent at any point (b tan θ, a sec θ) on 2nd hyperbolas is

$$\frac{y}{a}\sec\theta - \frac{x}{b}\tan\theta = 1 \qquad \text{...(2)}$$

If (1) and (2) are common tangents, then they should be identical therefore

$$\frac{\frac{\sec\phi}{a}}{\frac{-\tan\theta}{b}} = \frac{\frac{-\tan\phi}{b}}{\frac{\sec\theta}{a}} = \frac{1}{1} \Rightarrow \frac{\sec\theta}{a} = -\frac{\tan\phi}{b}$$

$$\text{Or } \sec\theta = -\frac{a}{b}\tan\phi \qquad \text{...(3)}$$

$$\text{and } -\frac{\tan\theta}{b} = \frac{\sec\phi}{a} \quad \text{or } \tan\theta = -\frac{b}{a}\sec\phi \qquad \text{...(4)}$$

$\because \sec^2\theta - \tan^2\theta = 1$

$$\Rightarrow \frac{a^2}{b^2}\tan^2\phi - \frac{b^2}{a^2}\sec^2\phi = 1 \qquad \{\text{from (3) and (4)}\}$$

or $\frac{a^2}{b^2}\tan^2\phi - \frac{b^2}{a^2}\tan^2\phi = 1 + \frac{b^2}{a^2}$ or $\left(\frac{a^2}{b^2} - \frac{b^2}{a^2}\right)\tan^2\phi = 1 + \frac{b^2}{a^2}$

$\therefore \quad \tan^2\phi = \frac{b^2}{a^2 - b^2}$ and $\sec^2\phi = 1 + \tan^2\phi = \frac{a^2}{a^2 - b^2}$

Hence the points of contact are

$\left\{\pm\frac{a^2}{\sqrt{(a^2 - b^2)}}, \pm\frac{b^2}{\sqrt{(a^2 - b^2)}}\right\}$ and $\left\{\pm\frac{b^2}{\sqrt{(a^2 - b^2)}}, \pm\frac{a^2}{\sqrt{(a^2 - b^2)}}\right\}$ from (3) and (4)

Length of common tangent i.e., the distance between the above points is

$\sqrt{2}\frac{(a^2 + b^2)}{\sqrt{(a^2 + b^2)}}$ and equation of common tangent on putting the values of secθ

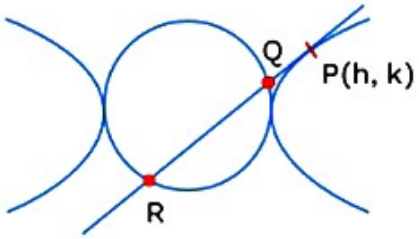
and tanθ in (1) is

$\pm\frac{x}{\sqrt{(a^2 + b^2)}} \mp \frac{y}{\sqrt{(a^2 - b^2)}} = 1$ or $x \mp y = \pm\sqrt{(a^2 - b^2)}$.

Q. **If the tangent at the point P(h, k) on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cuts the circle $x^2 + y^2 = a^2$ at the points $Q(x_1, y_1)$ and $R(x_2, y_2)$ then $\frac{1}{y_1} + \frac{1}{y_2}$ is equal to**

(A) $\frac{2}{k}$ **(B)** $\frac{1}{k}$ **(C)** $\frac{a}{k}$ **(D)** $\frac{b}{k}$

Sol. **(A)**



Equation of tangent at P is $\frac{xh}{a^2} - \frac{yk}{b^2} = 1$

Put $x = \left(1 + \frac{yk}{b^2}\right)\frac{a^2}{h}$ in equation $x^2 + y^2 = a^2$

$\left(1 + \frac{y^2k^2}{b^4} + \frac{2yk}{b^2}\right)\frac{a^4}{h^2} + y^2 = a^2$

$y^2\left(\frac{k^2a^4}{b^4h^2} + 1\right) + \frac{2ka^4}{b^2h^2}y + \left(\frac{a^4}{h^2} - a^2\right) = 0 \begin{matrix} \nearrow y_1 \\ \searrow y_2 \end{matrix}$

$$\frac{1}{y_1}+\frac{1}{y_2}=\frac{\frac{-2ka^4}{b^2h^2}}{\frac{a^4-a^2h^2}{h^2}}$$

$$=-\frac{2ka^2}{b^2\left(a^2-h^2\right)}=-\frac{2k}{b^2\left(1-\frac{h^2}{a^2}\right)}=-\frac{2k}{b^2\left(-\frac{k^2}{b^2}\right)}=\frac{2}{k}$$

Q. **Prove that two tangents drawn from any point on hyperbola $x^2-y^2=a^2-b^2$ to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ make complementary angle with the axis.**

Sol. Slopes of tangent an ellipse $\tan\theta$, $\cot\theta$

$m, \frac{1}{m}$

$$y=mx+\sqrt{a^2m^2+b^2} \quad ...(1)$$

$$y=\frac{x}{m}+\sqrt{\frac{a^2}{m^2}+b^2} \quad ...(2)$$

Locus of point of intersection of tangents

(2) tangent can be re-written as

$$my=x\pm\sqrt{a^2+b^2m^2}$$

$$my-x=\sqrt{a^2+b^2m^2} \quad ...(3)$$

(1) can be re-written as

$$y-mx=\sqrt{a^2m^2+b^2} \quad ...(4)$$

$(3)^2-(4)^2$

$$m^2y^2+x^2-2myx-y^2-m^2x^2+2my$$

$$=a^2\left(1-m^2\right)-b^2\left(1-m^2\right)$$

$$\left(x^2-y^2\right)\left(1-m^2\right)=\left(a^2-b^2\right)\left(1-m^2\right)$$

$$x^2-y^2=a^2-b^2$$

Which is given Hyperbola.

Comprehension for Q.1 to Q.2

The circle $x^2 + y^2 - 8x = 0$ and hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ intersect at the points A and B.

Q1 **Equation of a common tangent with positive slope to the circle as well as to the hyperbola is**

(A) $2x - \sqrt{5y} - 20 = 0$ **(B)** $2x - \sqrt{5} + 4 = 0$

(C) $3x - 4y + 8 = 0$ **(D)** $4x - 3y + 4 = 0$

Sol. **(B)**

$$x^2 - 8x + \left(\frac{x^2}{9} - 1\right)4 = 0$$

$\Rightarrow$ $13x^2 - 7x - 36 = 0$

$\Rightarrow$ $(13x + 6)(x - 6) = 0$

$\Rightarrow$ $A \equiv (6, 2\sqrt{3}), B \equiv (6, -2\sqrt{3})$

Let tangent to $\frac{x^2}{9} - \frac{y^2}{4} = 1$ be

$$y = mx \pm \sqrt{9m^2 - 4} \quad \ldots(1)$$

$\therefore$ (1) is tangent to circle

$$\Rightarrow \frac{4m \pm \sqrt{9m^2 - 4}}{\sqrt{1 + m^2}} = 4$$

$$\Rightarrow 16m^2 + \left(9m^2 - 4\right) \pm 8\sqrt{9m^2 - 4}$$

$$= 16 + 16m^2$$

$$\Rightarrow 495m^4 + 104m^2 - 400 = 0$$

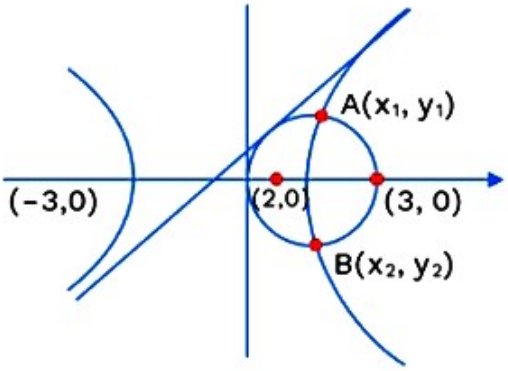
$$\Rightarrow m^2 = \frac{-104 + \sqrt{(104) + 1600(495)}}{2(495)}$$

$$\Rightarrow m^2 = \frac{4}{5} \Rightarrow m = \frac{2}{\sqrt{5}}$$

Equation of tangent is

$$y = \frac{2x}{\sqrt{5}} + \frac{4}{5}$$

$$\Rightarrow \sqrt{5}y = 2x + 4$$



Q2 **Equation of the circle with AB as its diameter is**

(A) $x^2 + y^2 - 12x + 24 = 0$ **(B)** $x^2 + y^2 + 12x + 24 = 0$

(C) $x^2 + y^2 + 24 - 12 = 0$ **(D)** $x^2 + y^2 - 24x - 12 = 0$

Sol. **(A)**

$(x-6)^2 + (y+2\sqrt{3})(y-2\sqrt{3}) = 0$

$\Rightarrow x^2 + y^2 - 12x + 4 = 0$

Q. **Find its range of eccentricity if it is possible to draw the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ of slope 2.**

Sol. Tangent having slope m is $y = mx \pm \sqrt{(a^2m^2 - b^2)}$

The tangent having slope 2 is $y = 2x \pm \sqrt{(4a^2 - b^2)}$,

which is real if

$4a^2 - b^2 \geq 0$

or $\frac{b^2}{a^2} \leq 4$ or $e^2 - 1 \leq 4$ or $e^2 \leq 5$

or $1 < e \leq \sqrt{5}$ (for hyperbola $e > 1$)

EQUATION OF NORMALS

(1) Point form:

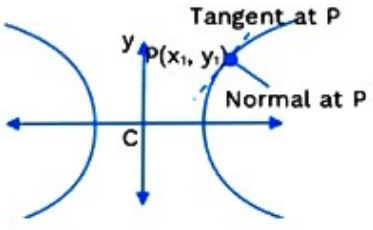
Equation of normal to a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at a point (x_1, y_1) is

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2$$

(2) Parametric form:

The equation of normal at $(a \sec \phi, b \tan \phi)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$ax\cos\phi + by\cot\phi = a^2 + b^2$$



Illustrations :

Q. **Prove that the line lx + my – n = 0 will be a normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ if $\frac{a^2}{l^2}-\frac{b^2}{m^2}=\frac{\left(a^2+b^2\right)^2}{n^2}$**

Sol. The equation of any normal to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

$ax\cos\phi+by\cot\phi=a^2+b^2$ or $ax\cos\phi+by\cot\phi-\left(a^2+b^2\right)=0$ ...(1)

Since the straight line lx+my–n=0 is given normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

then (1) and lx+my–n=0 represent the same line

$$\frac{a\cos\phi}{l}=\frac{b\cot\phi}{m}=\frac{\left(a^2+b^2\right)}{n}$$

or $\sec\phi=\frac{na}{l\left(a^2+b^2\right)}$ and $\tan\phi=\frac{nb}{m\left(a^2+b^2\right)}$

$\because\ \sec^2\phi-\tan^2\phi=1$

$$\therefore\ \frac{n^2a^2}{l^2\left(a^2+b^2\right)^2}-\frac{n^2b^2}{m^2\left(a^2+b^2\right)^2}=1$$

$$\Rightarrow\ \frac{a^2}{l^2}-\frac{b^2}{m^2}=\frac{\left(a^2+b^2\right)^2}{n^2}$$

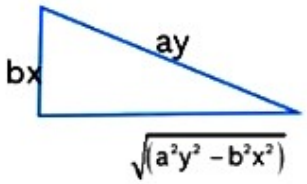
Q. **Find the locus of the foot of perpendicular from the centre upon any normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.**

Sol. Normal at $P(a\sec\phi, b\tan\phi)$ is

$ax\cos\phi+by\cot\phi=a^2+b^2$ ...(1)

and equation of line perpendicular to (1) and passing through origin is

$bx-ay\sin\phi=0$ ...(2)

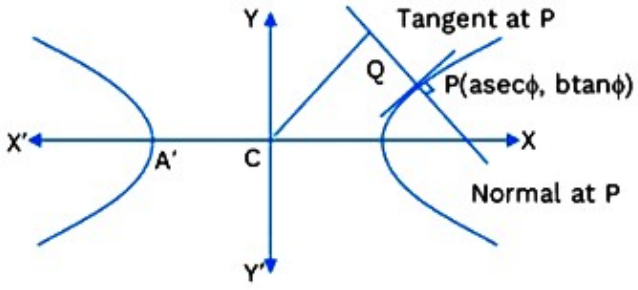


From (2)

$$\sin\phi = \frac{bx}{ay}$$

$$\therefore \quad \cos\phi = \frac{\sqrt{\left(a^2y^2 - b^2x^2\right)}}{ay}$$

$$\text{and} \quad \cot\phi = \frac{\sqrt{\left(a^2y^2 - b^2x^2\right)}}{bx}$$



Now by putting value of cos ϕ and cot ϕ in equation (1), ϕ will be eliminated and get the locus.

$$\therefore \quad \text{from (I)}, ax \times \frac{\sqrt{a^2y^2 - b^2x^2}}{ay} + by \times \frac{\sqrt{\left(a^2y^2 - b^2x^2\right)}}{bx} = a^2 + b^2$$

$$\Rightarrow \quad \left(x^2 + y^2\right)\sqrt{\left(a^2y^2 - b^2x^2\right)} = \left(a^2 + b^2\right)xy$$

$$\text{or} \quad \left(x^2 + y^2\right)^2\left(a^2y^2 - b^2x^2\right) = \left(a^2 + b^2\right)^2 x^2y^2$$

which is required locus.

Q. **Prove that the part of the tangent at any point of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ intercepted between the point of contact and the transverse axis is a harmonic mean between the lengths of the perpendicular drawn from the foci on the normal at the same point.**

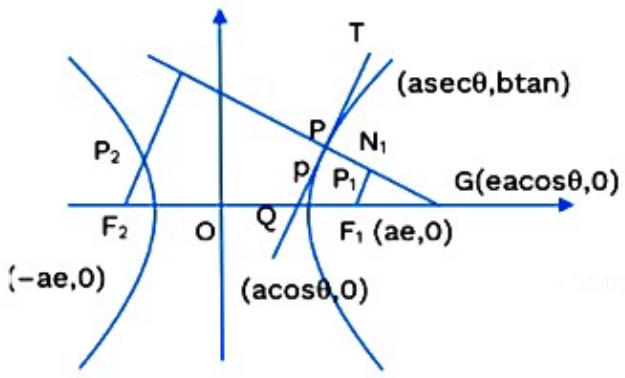
Sol. Tangent at point $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$;

Normal at point P is $\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2e^2$

From the diagram $\frac{p}{p_1} = \frac{QG}{F_1G}$

$$= \frac{e^2a\sec\theta - a\cos\theta}{e^2a\sec\theta - ae} = \frac{e^2 - \cos^2\theta}{e^2 - e\cos\theta}$$

$$= \frac{e + \cos\theta}{e}$$



$$\Rightarrow \frac{p}{p_1} = 1 + \frac{\cos\theta}{e}$$

Similarly $\frac{p}{p_2} = 1 - \frac{\cos\theta}{e}$

$$\Rightarrow \frac{p}{p_1} + \frac{p}{p_2} = 2$$

$$\Rightarrow \frac{2}{p} = \frac{1}{p_1} + \frac{1}{p_2}$$

Q. **Normals at points θ_1 and θ_2 are drawn to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ meeting the conjugate axis at G_1 and G_2, respectively. If θ_1 and θ_2 are complementary angles, and if $CG_1 \cdot CG_2 = \frac{a^2e^k}{e^2-1}$ where 'k' is natural number and C is the centre of the hyperbola and e is its eccentricity. Find value of 'k'**

Sol. Normal at point P($a\sec\theta_1$, $b\tan\theta_1$) is $ax\cos\theta_1 + by\cot\theta_1 = (a^2 + b^2)$

It meets the conjugate axis at $G_1\left(0, \frac{a^2+b^2}{b}\tan\theta_1\right)$.

Normal at point Q($a\sec\theta_2$, $b\tan\theta_2$) is $ax\cos\theta_2 + by\cot\theta_2 = (a^2 + b^2)$.

It meets the conjugate axis at $G_2\left(0, \frac{a^2+b^2}{b}\tan\theta_2\right)$

$$CG_1 \cdot CG_2 = \frac{\left(a^2+b^2\right)^2}{b^2}\tan\theta_1\tan\theta_2$$

$$= \frac{\left(a^2+b^2\right)^2}{b^2} \left(\because \theta_1 + \theta_2 = \frac{\pi}{2}\right)$$

$$= \frac{a^4\left(1+\frac{b^2}{a^2}\right)^2}{b^2}$$

$$= \frac{a^2e^4}{e^2-1}$$

Hence k=4

PROPERTIES :

(1) Normal and tangent at any point $P(x_1, y_1)$ meet the transverse axis at G and T respectively.

The equation of normal at P on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2 \qquad ...(1)$$

The normal (1) meets the x-axis i.e. then put y = 0 in (1) we will get co-ordinates of G

i.e. $G\left(\frac{(a^2+b^2)}{a^2}x_1, 0\right)$ or $(e^2x_1, 0)$

$\therefore$ $CG = e^2x_1$

Now

$SG = CG - CS = e^2x_1 - ae = e(ex_1 - a) = e \cdot SP$

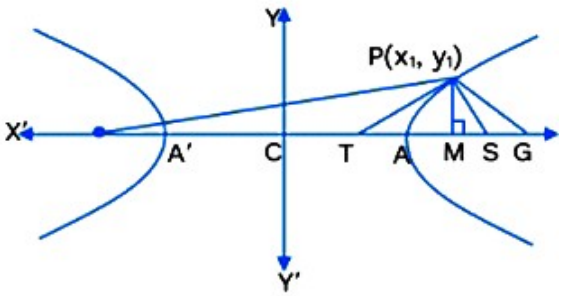
Similarly $S'G = e \cdot S'P$

$\therefore$ $\frac{SG}{S'G} = \frac{SP}{S'P}$

This relation shows that the normal PG is the external bisector of the angle SPS'. The tangent PT (perpendicular to PG) is therefore the internal bisector of the angle SPS'.

The tangent & normal at any point of a hyperbola bisect the angle between the focal radii.

(2) The tangent and normal at any point of a hyperbola bisect the angle between the focal radii. This spells the reflection property of the hyperbola as "**An incoming light ray** "aimed towards one focus is reflected from the outer surface of the hyperbola towards the other focus.

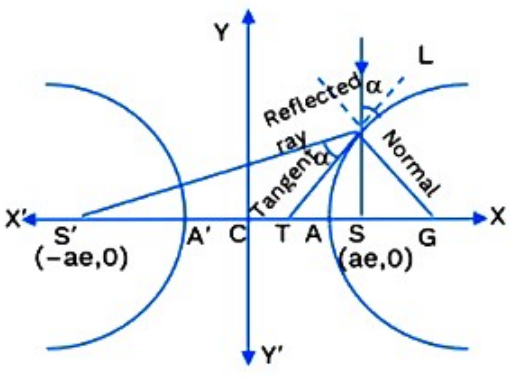


Point to Remember!!!

That the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the hyperbola

$$\frac{x^2}{a^2 - k^2} - \frac{y^2}{k^2 - b^2} = 1 (a > k > b > 0)$$

are co-focal and therefore orthogonal.



It follows that if an ellipse and a hyperbola have the same foci, they cut at right angles at any of their common point.

(3) Locus of the feet of the perpendicular drawn from focus of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ upon any tangent is its auxiliary circle & the product of the feet of these perpendiculars is b^2.

i.e., square of semi-conjugate axis.

Similar property exists in ellipse also.

(4) The portion of the tangent between the point of contact & the directrix subtends a right angle at the corresponding focus.

(5) The foci of the hyperbola and the points P and Q in which any tangent meets the tangents at the vertices are concyclic with PQ as diameter of the circle.

Equation of Chord Joining Points P(α) and Q(β)

Equation of chord passing through the points $P(a\sec\alpha,, b\tan\alpha)$ and $Q(a\sec\beta, b\tan\beta)$ is given by

$$\begin{vmatrix} x & y & 1 \\ a\sec\alpha & b\tan\alpha & 1 \\ a\sec\beta & b\tan\beta & 1 \end{vmatrix} = 0.$$

$\Rightarrow bx(\tan\alpha-\tan\beta)- ay(\sec\alpha - \sec\beta) + ab(\sec\alpha\tan\beta - \sec\beta\tan\alpha) = 0$

$\Rightarrow bx\sin(\alpha - \beta) - ay(\cos\beta - \cos\alpha) + ab(\sin\beta - \sin\alpha) = 0$

$$bx\cos\left(\frac{\alpha-\beta}{2}\right) - ay\sin\left(\frac{\alpha+\beta}{2}\right) - ab\cos\left(\frac{\alpha+\beta}{2}\right) = 0$$

$$\frac{x}{a}\cos\left(\frac{\alpha-\beta}{2}\right) - \frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha+\beta}{2}\right)$$

Q. **If (asecθ, btanθ) and (asecϕ, b tanϕ)are the ends of a focal chord of the**

Prove that $\tan\frac{x}{a}\tan\frac{\phi}{2}=\frac{1-e}{1+e}$

Sol. The equation of the chord joining

$(a\sec\theta, b\tan\theta)$ and $(a\sec\phi, b\tan\phi)$ is

$$\frac{x}{a}\cos\frac{\theta-\phi}{2}-\frac{y}{b}\sin\frac{\theta+\phi}{2}=\cos\frac{\theta+\phi}{2}$$

This passes through (ae, 0)

$$e\cos\left(\frac{\theta-\phi}{2}\right)=\cos\left(\frac{\theta+\phi}{2}\right)$$

$$e=\frac{\cos\left(\frac{\theta+\phi}{2}\right)}{\cos\left(\frac{\theta-\phi}{2}\right)}$$

$$\Rightarrow\frac{e-1}{e+1}=\frac{\cos\left(\frac{\theta+\phi}{2}\right)-\cos\left(\frac{\theta-\phi}{2}\right)}{\cos\left(\frac{\theta+\phi}{2}\right)+\cos\left(\frac{\theta-\phi}{2}\right)}$$

$$\Rightarrow\quad\tan\frac{\theta}{2}\tan\frac{\phi}{2}=\frac{1-e}{1+e}$$

PAIR OF TANGENTS

The combined equation of the pair of tangents drawn from a point $P(x_1, y_1)$ lying outside the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$$\left(\frac{x^2}{a^2} - \frac{y^2}{b^2} - 1\right)\left(\frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1\right) = \left(\frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1\right)^2 \text{ or } SS_1 = T^2$$

where $S = \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1$; $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$ and $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

DIRECTOR CIRCLE

The locus of the point of intersection of the tangents to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, which is perpendicular to each other, is called the director circle.

Any tangent to hyperbola is

$$y = mx + \sqrt{a^2m^2 - b^2} \qquad ...(1)$$

If it passes through P(h, k) then

$$k - mh = \sqrt{a^2m^2 - b^2}$$

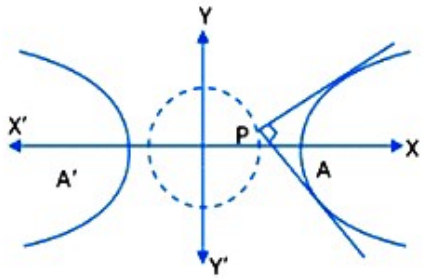
$$k^2 + m^2h^2 - 2mk = a^2m^2 - b^2$$

$$m^2\left(h^2 - a^2\right) - 2mhk + \left(k^2 + b^2\right) = 0$$

It is quadratic in m therefore it has two roots m_1 and m_2. Hence two tangents (real or imaginary) can be drawn from P(h, k).

If pair of perpendicular tangents are drawn from P(h, k) then $m_1m_2 = \frac{k^2 + b^2}{h^2 - a^2} = -1$

Locus of P(h, k) is $x^2 + y^2 = a^2 - b^2$



Point to Remember!!!

(i) If $b^2 < a^2$ this circle is real

(ii) If $b^2 = a^2$ the radius of the circle is zero & it reduces to a point circle at the origin. In this case the centre is the only point from which the tangents at right angles can be drawn to the curve.

(iii) If $b^2 > a^2$, the radius of the circle is imaginary, so that there is no such circle & so no tangents at right angle can be drawn to the curve.

Q. **Find the equation of tangents and angle between them that can be drawn from the point (4, 3) to the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$?**

Sol. Given point $P \equiv (4, 3)$

Hyperbola $S \equiv \frac{x^2}{16}-\frac{y^2}{9}-1$

Equation of pair of tangents is $SS_1 = T^2$

$$\Rightarrow\left(\frac{x^2}{16}-\frac{y^2}{9}-1\right)(-1)=\left(\frac{4x}{16}-\frac{3y}{9}-1\right)^2$$

$$\Rightarrow -\frac{x^2}{16}+\frac{y^2}{9}+1=\frac{x^2}{16}+\frac{y^2}{9}+1-\frac{xy}{6}-\frac{x}{2}+\frac{2y}{3}$$

$$\Rightarrow \frac{x^2}{8}-\frac{xy}{6}-\frac{x}{2}+\frac{2y}{3}=0$$

$$\Rightarrow 3x^2 - 4xy - 12x + 16y = 0$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$$

Q. **If major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to 2R and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola, then the correct relation is**

(A) $4e_1^2 - e_2^3 = 6$ **(B)** $e_1^2 - 4e_2^2 = 2$ **(C)** $4e_2^2 - e_1^2 = 6$ **(D)** $2e_1^3 - e_2^2 = 4$

Sol. **(C)**

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, \quad \frac{x^2}{a^2}-\frac{y^2}{B^2}=1$$

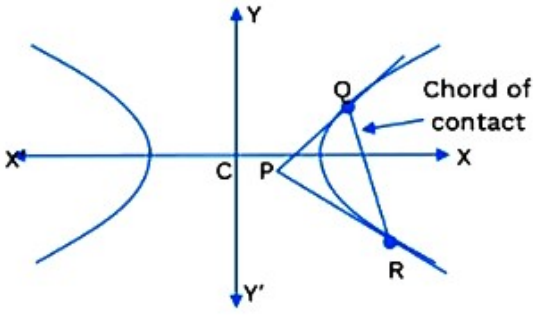
$$a^2+b^2=4R^2=a^2\left(2-e_1^2\right)$$

$$a^2-B^2=R^2=a^2\left(2-e_2^2\right)$$

$$4=\frac{2-e_1^2}{2-e_2^2} \Rightarrow 4e_2^2 - e_1^2 = 6$$

CHORD OF CONTACT

If the tangents from a point $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ touch the hyperbola at Q and R, then the equation of the chord of contact QR is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$



Illustrations :

Q. **If tangent to the parabola $y^2 = 4ax$ intersect the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at A and B, then find the locus of point of intersection of tangents at A and B.**

Sol. Let P = (h, k) be the point of intersection of tangents at A and B.

Therefore, the equation of chord of contact AB is of hyperbola is $\frac{xh}{a^2} - \frac{yk}{b^2} = 1$

Or $y = \frac{xb^2h}{ka^2} - \frac{b^2}{k}$ touches the given parabola

Therefore $-\frac{b^2}{k} = \frac{a}{\left(\frac{b^2h}{a^2k}\right)} \Rightarrow$ Locus of point P(h, k) is $y^2 = -\frac{b^4}{a^3}x$

Q. **Tangents drawn from a point P to the parabola $y^2 = 4ax$ are so that the chord of contact is a tangent to the hyperbola $x^2 - y^2 = a^2$ .Show that the locus of P is an ellipse.**

Sol. Let the co-ordinates of P are (h, k).

Equation of the chord of contact with respect to the parabola $y^2 = 4ax$ is

$ky = 2a(x + h)$ or $y = \frac{2a}{k}x + \frac{2ah}{k}$

It is a tangent to the hyperbola $x^2 - y^2 = a^2$, so we have

$\left(\frac{2ah}{k}\right)^2 = a^2\left(\frac{2a}{k}\right)^2 - a^2$ or $\frac{4h^2}{k^2} = \frac{4a^2}{k^2} - 1$

or $4h^2 + k^2 = 4a^2$

The locus of P is therefore $4x^2 + y^2 = 4a^2$

Which is clearly an ellipse.

Equation of the chord bisected at a given point

The equation of the chord of the hyperbola

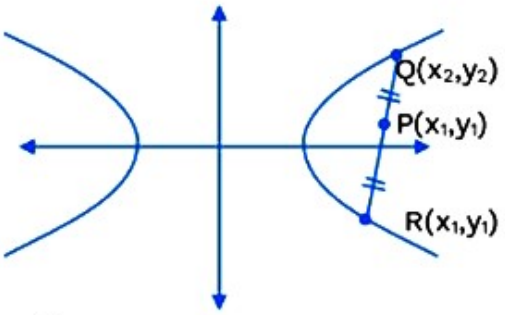
$$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1,$$

bisected at the point $P(x_1,y_1)$ is

$$\frac{xx_1}{a^2}-\frac{yy_1}{b^2}-1=\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1$$

or $T = S_1$, where $T=\frac{xx_1}{a^2}-\frac{yy_1}{b^2}-1$

$$S_1=\frac{x_1^2}{a^2}-\frac{y_1^2}{b^2}-1$$



Illustrations :

Q. **Find the equation to the locus of the middle points of the chords of the hyperbola $2x^2-3y^2=1$, each of which makes an angle of 45° with the x-axis.**

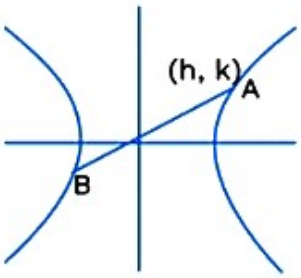
Sol. Equation of chord AB with $T = S_1$ is given by

$$\frac{hx}{3}-\frac{ky}{2}=\frac{h^2}{3}-\frac{k^2}{2}$$

Given that it has slope = tan 45°.

Hence $\frac{2h}{3k}=1$

2x =3y (as which is the required locus)



Q. **Find the locus of the mid points of the chords of the circle $x^2+y^2=16$, which are tangent to the hyperbola $9x^2-16y^2=144$.**

Sol. Any tangent to hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ is $y=mx+\sqrt{(16m^2-9)}$ ...(1)

Let (x_1, y_1) be the mid-point of the chord of the circle $x^2 + y^2 = 16$, then equation of the chord is $(T = S_1)$.

$xx_1+yy_1-\left(x_1^2+y_1^2\right)=0$ ...(2)

Since (1) and (2) are same, therefore

We get $\frac{m}{x_1} = -\frac{1}{y_1} = \frac{\sqrt{16m^2 - 9}}{-\left(x_1^2 + y_1^2\right)} \Rightarrow m = -\frac{x_1}{y_1}$ and $\left(x_1^2 + y_1^2\right)^2 = y_1^2\left(16m^2 - 9\right)$

Eliminating m and replacing $\left(x_1, y_1\right)$ by (x, y), we get the required locus is

$\left(x^2 + y^2\right)^2 = 16x^2 - 9y^2$.

Q. **A tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ cuts the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at P and Q.**

Show that the locus of the mid point of PQ is $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = \frac{x^2}{a^2} - \frac{y^2}{b^2}$

Sol. Tangent to hyperbola at P(θ) is

$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$ ...(i)

Also the chord of the ellipse with middle point (h, k) is

$\frac{hx}{a^2} + \frac{ky}{b^2} = \frac{h^2}{a^2} + \frac{k^2}{b^2} = \lambda$ (say) ...(ii)

Comparing (i) and (ii), we get

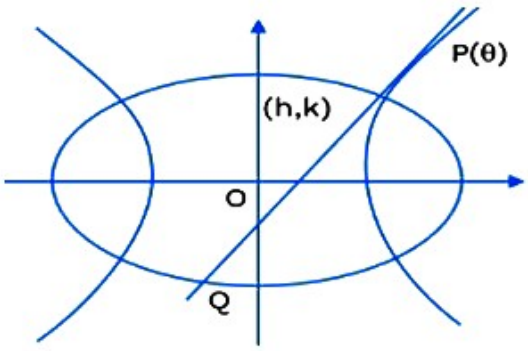
$\frac{\sec\theta}{a}\frac{a^2}{h} = -\frac{\tan\theta}{b}\frac{b^2}{k} = \frac{1}{\lambda}$

$\Rightarrow \sec\theta = \frac{h}{\lambda a}$ and $\tan\theta = -\frac{k}{\lambda b}$

$\therefore \frac{h^2}{\lambda^2 a^2} - \frac{k^2}{\lambda^2 b^2} = 1$(Eliminating θ)

$\Rightarrow \frac{h^2}{a^2} - \frac{k^2}{b^2} = \lambda^2 = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2}\right)^2$

Hence locus is $\left(\frac{x^2}{a^2} + \frac{y^2}{b^2}\right)^2 = \frac{x^2}{a^2} - \frac{y^2}{b^2}$



Q. **Show that the mid points of focal chords of a hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ lie on another similar hyperbola.**

Sol. Equation of chord AB with $T = S_1$

$$\frac{hx}{a^2} - \frac{ky}{b^2} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

If passes through (ae, 0)

$$\Rightarrow \quad \frac{he}{a} = \frac{h^2}{a^2} - \frac{k^2}{b^2}$$

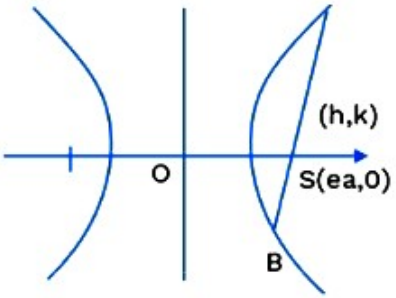
Locus is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{ex}{a}$

$$\Rightarrow \quad \frac{1}{a^2}\left[x^2 - aex\right] - \frac{y^2}{b^2} = 0$$

$$\Rightarrow \quad \frac{\left(x - \frac{ae}{2}\right)^2 - \frac{a^2e^2}{4}}{a^2} - \frac{y^2}{b^2} = 0$$

$$\Rightarrow \frac{\left(x - \frac{ae}{2}\right)^2}{a^2} - \frac{y^2}{b^2} = \frac{e^2}{4}$$

which is also a hyperbola with eccentricity e.



ASYMPTOTES OF HYERPBOLA

If the length of the perpendicular let fall from a point on a hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called the Asymptote of the Hyperbola.

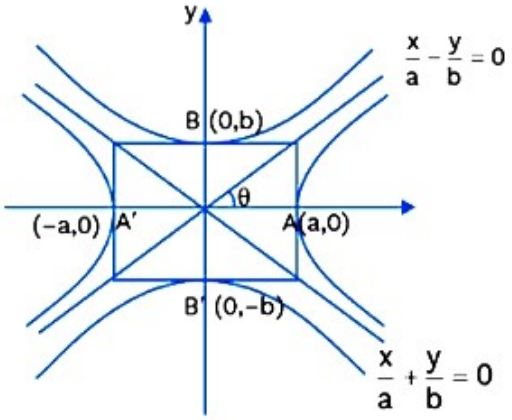
Let $y = mx + c$ is the asymptote of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Solving these two we get the quadratic as

$$\left(b^2 - a^2m^2\right)x^2 - 2a^2mcx - a^2\left(b^2 + c^2\right) = 0 \quad ...(1)$$

In order that $y=m\ x+c$ be an asymptote, both roots of equation (1) must approach at infinity, the conditions for which are coeff of $x^2 = 0$ coeff of $x = 0$

$$\Rightarrow \quad b^2 - a^2m^2 = 0 \text{ or } m = \pm\frac{b}{a} \quad \&$$



Know the facts

In short asymptote is tangent to hyperbola at infinity.

$a^2mc = 0 \Rightarrow c = 0$

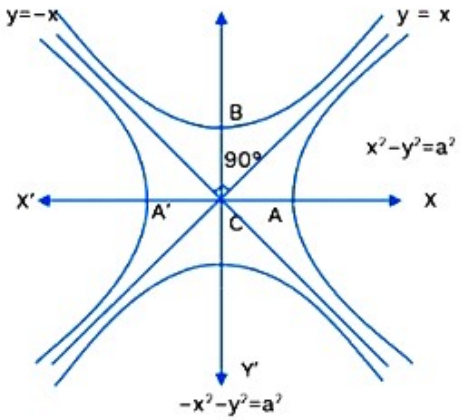
equations of asymptote are

$\frac{x}{a} + \frac{y}{b} = 0$ and $\frac{x}{a} - \frac{y}{b} = 0$

combined equation to the asymptotes

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 0$$

As asymptotes of any hyperbola or a curve is a straight line which touches in it two points at infinity.



Note :

(i) If b = a then $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ reduces $x^2 - y^2 = a^2$

The asymptotes of rectangular hyperbola $x^2 - y^2 = a^2$ are $y = \pm x$ which are at right angles.

(ii) Let

$$H \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1 = 0;$$

$$A \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ and } C = \frac{x^2}{a^2} - \frac{y^2}{b^2} + 1 = 0$$

be the equation of the hyperbola, asymptotes and the conjugate hyperbola respectively, then clearly C + H =2A i.e., all the equation different only by constant term and the constant term of H, A and C are in A.P.

(iii) The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis. Here area of rectangle is 4ab.

Know the facts

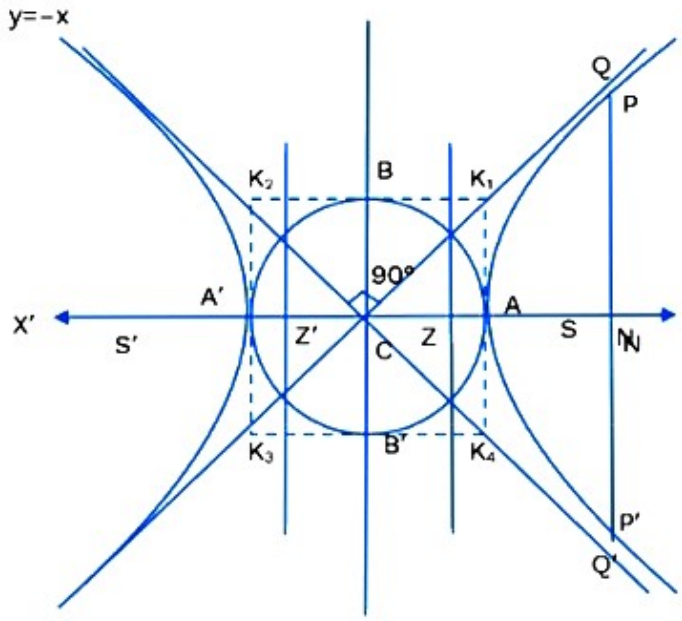
(i) A hyperbola and its conjugate have the same asymptote.

(ii) The asymptotes pass through the centre of the hyperbola & the bisectors of the angles between the asymptotes are the axes of the hyperbola.

(iii) Asymptotes are the tangent to the hyperbola from the centre.

PROPERTIES OF ASYMPTOTES

(i) Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.



(ii) If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point & the curve is always equal to the square of the semi conjugate axis.

(iii) The tangent at any point P on a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ with centre C, meets the asymptotes in Q and R and cuts off a ΔCQR of constant area equal to ab from the asymptotes & the portion of the tangent intercepted between the asymptote is bisected at the point of contact.

Point to Remember!!!

A simple method to find the coordinates of the centre of the hyperbola is expressed as a general equation of degree 2.

i.e. let f(x,y) represents a hyperbola.

Find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$. Then the point of intersection of $\frac{\partial f}{\partial x}=0$ & $\frac{\partial f}{\partial y}=0$ gives the centre of the hyperbola.

Point to Remember!!!

The angle between the asymptotes of $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is $2\tan^{-1}\left(\frac{b}{a}\right)$ and if the angle between the asymptote of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is 2θ then $e=\sec\theta$.

Illustrations :

Q. **Find the asymptotes of the hyperbola $xy - 3y - 2x = 0$.**

Sol. Since equation of a hyperbola and its asymptotes differ in constant terms only.
$\therefore$ Pair of asymptotes is given by $xy - 3y - 2x + \lambda = 0$ ...(1)
above equation represents two straight lines.
$\therefore \quad abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$$\Rightarrow \quad 0 + 2 \times \frac{-3}{2} \times -1 \times \frac{1}{2} - 0 - 0 - \lambda\left(\frac{1}{2}\right)^2 = 0 \Rightarrow \lambda = 6$$

From (1), the asymptotes of given hyperbola are given by
$xy - 3y - 2x + 6 = 0$ or $(y - 2)(x - 3) = 0$.
$\therefore$ Asymptotes are $x - 3 = 0$ and $y - 2 = 0$

Q. **The asymptotes of a hyperbola having centre at the point (1, 2) are parallel to the lines $2x + 3y = 0$ and $3x + 2y = 0$. If the hyperbola passes through the point (5,3), show that its equation is $(2x + 3y - 8)(3x + 2y + 7) = 154$.**

Sol. Let the asymptotes be $2x + 3y + \lambda = 0$ and $3x + 2y + \mu = 0$. Since asymptotes passes through (1, 2), then $\lambda = -8$ and $\mu = -7$.
Thus the equation of asymptotes are $2x + 3y - 8 = 0$ and $3x + 2y - 7 = 0$.
Let the equation of hyperbola be $(2x + 3y - 8)(3x + 2y - 7) + \lambda = 0$ ...(1)
Hyperbola passes through (5,3) then $(10 + 9 - 8)(15 + 6 - 7) + \lambda = 0$
$\therefore \lambda = -154$.
Putting the value of λ in (1), we get $(2x+3y-8)(3x+2y-7)-154=0$ which is the equation of required hyperbola.

Q. **Find the asymptotes of the hyperbola, $3x^2 - 5xy - 2y^2 - 5x + 11y - 8 = 0$. Also find the equation of the conjugate hyperbola and their centre.**

Sol. Let equation of asymptotes
$3x^2 - 5xy - 2y^2 - 5x + 11y + \lambda = 0$
for pair of lines
$\lambda = -12$
Let equation of conjugate hyperbola
$3x^2 - 5xy - 2y^2 - 5x + 11y + k = 0$
Now $C + H = 2A$
$C = 2A - H$; $k = -16$

Q. **A transversal cuts the same branch of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ in P, P' and the asymptotes in Q, Q' Prove that : PQ = P'Q'**

Sol. Hyperbola is $b^2x^2 - a^2y^2 = a^2b^2$

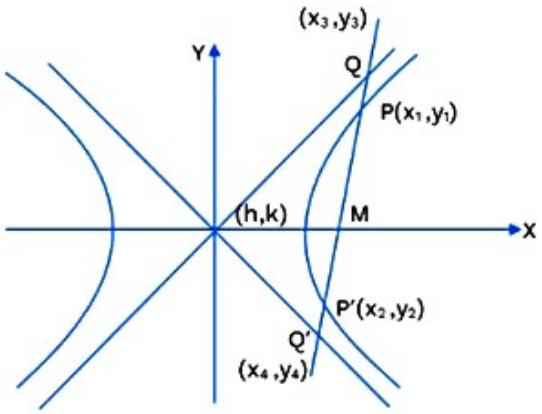
Let the transversal be $y = mx + c$

Solving hyperbola and transversal we get

$b^2x^2 - a^2(mx + c)^2 = a^2b^2$

$\Rightarrow \left(b^2 - a^2m^2\right)x^2 - 2a^2mcx - a^2\left(c^2 + b^2\right) = 0$

$\Rightarrow \frac{x_1 + x_2}{2} = \frac{2a^2mc}{b^2 - a^2m^2}$ . .. (i)



Solving y=mx+c with pair of asymptotes

$b^2x^2 - a^2y^2 = 0$ we have

$\left(b^2 - a^2m^2\right)x^2 - 2a^2mcx - a^2c^2 = 0$

$\Rightarrow \frac{x_3 + x_4}{2} = \frac{2a^2mc}{b^2 - a^2m^2}$ ...(ii)

From (i) & (ii)

$\Rightarrow$ MQ = MQ' and MP = MP'

$\Rightarrow$ PQ = P'Q'

Q. **Let S_1 and S_2 be the foci of a rectangular hyperbola, which has the centre at Q, then for any point P on the hyperbola $S_1P \cdot S_2P$ equals to**

(A) $S_1S_2^2$ **(B) QS_1^2** **(C) QP^2** **(D) $4QP^2$**

Sol. **(C)**

$(S_1P)\,(S_2P)$

$= e\left(a\sec\theta - \frac{a}{e}\right)e\left(a\sec\theta + \frac{a}{e}\right)$

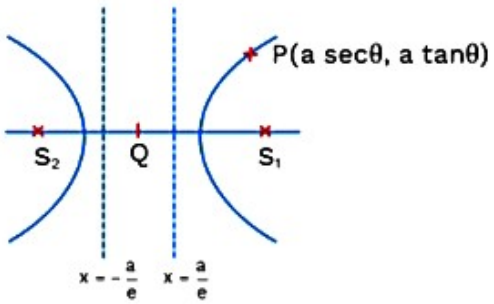
$= a^2\,(e^2\sec^2\theta - 1)$

$= a^2\,(2\sec^2\theta - 1)$

$= a^2\,(\sec^2\theta + \tan^2\theta)$

$= a^2\sec^2\theta + a^2\tan^2\theta$

$= QP^2$



Q. **The tangent at any point P of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ meets one of the asymptotes in Q and L, M are the feet of the perpendiculars from Q on the axes. Prove that LM passes through P.**

Sol. $P(a\sec\theta, b\tan\theta)$

Tangent of P is $\frac{x\sec\theta}{a}-\frac{y}{b}\tan\theta=1$ ...(1)

Let asymptote

$\frac{x}{a}-\frac{y}{b}=0$ ...(2)

Let Q be (h, k)

Q lie on line (1) and line (2)

$\frac{h}{a}-\frac{k}{b}=0$

$h=\frac{ka}{b}$

$\frac{k\sec\theta}{b}-\frac{k\tan\theta}{b}=1$

$k=\frac{b}{\sec\theta-\tan\theta}$

$h=\frac{a}{\sec\theta-\tan\theta}$

Co-ordinates of Q are $\left(\frac{a}{\sec\theta-\tan\theta},\frac{b}{\sec\theta-\tan\theta}\right)$

$L\left(\frac{a}{\sec\theta-\tan\theta},0\right)$

$m\left(0,\frac{b}{\sec\theta-\tan\theta}\right)$

Equation of LM $\frac{x(\sec\theta-\tan\theta)}{a}+\frac{y(\sec\theta-\tan\theta)}{b}=1$

Satisfying coordinates $P(a\sec\theta, b\tan\theta)$

$\sec^2\theta-\sec\theta\tan\theta+\sec\theta\tan\theta-\tan^2\theta=1$

Hence P lies on LM

Q. **A normal is drawn to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at P which meets the transverse axis (TA) at G If perpendicular from G on the asymptote meets it at L, show that LP is parallel to conjugate axis.**

Sol. Equation of GL with slope $-a/b$ and passing through $(e^2x_1,0)$ is

$$y-0=-\frac{a}{b}(x-e^2x_1)$$

$$ax+by=ae^2x_1$$

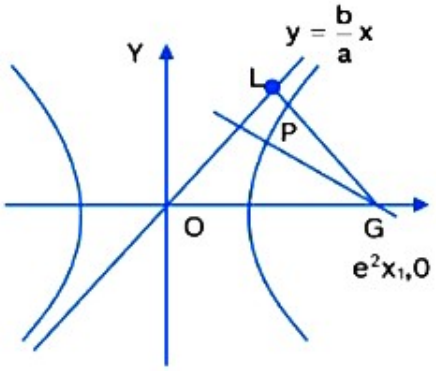
Put $y=\frac{b}{a}x$

$$ax+b\frac{b}{a}x=ae^2x_1$$

$$x\left[\frac{a^2+b^2}{a}\right]=ae^2x_1$$

$$x=x_1$$

Thus abscissa of point L is x_1 which is same as that of P
Hence, LP is parallel to conjugate axis.



The rectangular hyperbola $xy = c^2$

When the centre of any rectangular hyperbola be at the origin and its asymptotes coincide with the co-ordinate axes then equation of hyperbola is $xy = c^2$

Here the equation of asymptotes is $xy = 0$ and the equation conjugate hyperbola is $xy = -c^2$

Properties of rectangular hyperbola $xy = c^2$

(a) Equation of a chord joining the points $P(t_1)$ and $Q(t_2)$ is $x + t_1t_2y = c(t_1+t_2)$ with slope $m=-\frac{1}{t_1t_2}$.

(b) Equation of the tangent at $P(x_1, y_1)$ is $\frac{x}{x_1}+\frac{y}{y_1}=2$ and at $P(t)$ is $\frac{x}{t}+ty=2c$.

Point to Remember!!!

(i) The equation of a rectangular hyperbola having co-ordinate axes as its asymptotes is $xy = c^2$. If the asymptotes of a rectangular hyperbola are $x = \alpha, y = \beta$ then its equation is $(x-\alpha)(y-\beta)=c^2$ or $xy-\alpha y-\beta x+\lambda=0$.

(ii) Parametric equation of $xy = c^2$ is $x = ct$ and $y=\frac{c}{t}$.

$\therefore (x,y)=\left(ct,\frac{c}{t}\right)(t\neq 0)$ is called a 't' point on the rectangular hyperbola.

(c) Point of intersection of tangents a 't_1' and 't_2' is $\left(\frac{2ct_1t_2}{t_1+t_2}, \frac{2c}{t_1+t_2}\right)$

(d) Equation of normal is $y - \frac{c}{t} = t^2(x - ct)$ or $xt^3 - yt - ct^4 + c = 0$

Illustrations :

Q. **If the normal at the point 't', to the rectangular hyperbola $xy = c^2$ meets it again at the point 't_2' prove that $t_1^3t_2 = -1$.**

Sol. Since the equation of normal at $\left(ct_1, \frac{c}{t_1}\right)$ to the hyperbola $xy = c^2$ is $xt_1^3 - yt_1 - ct_1^4 + c = 0$.

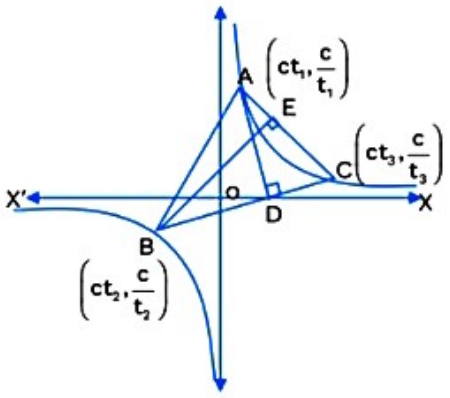
but this passes through $\left(ct_2, \frac{c}{t_2}\right)$ then $ct_2t_1^3 - \frac{c}{t_2}t_1 - ct_1^4 + c = 0$

$\Rightarrow t_2^2t_1^3 - t_1 - t_1^4t_2 + t_2 = 0 \Rightarrow t_2t_1^3(t_2 - t_1) + (t_2 - t_1) = 0$

$\Rightarrow (t_1^3t_2 + 1)(t_2 - t_1) = 0 \Rightarrow t_1^3t_2 + 1 = 0 \quad [\because t_2 \neq t_1] \quad \therefore t_1^3t_2 = -1$

Q. **A triangle has its vertices on a rectangular hyperbola. Prove that the orthocentre of the triangle also lies on the same hyperbola.**

Sol. $\therefore$ Co-ordinates of A, B and C $\left(ct_1, \frac{c}{t_1}\right), \left(ct_2, \frac{c}{t_2}\right)$ and $\left(ct_3, \frac{c}{t_3}\right)$ are respectively are the vertices of a triangle lie on the rectangular hyperbola $xy = c^2$



Now slope of BC is $\frac{\frac{c}{t_3} - \frac{c}{t_2}}{ct_3 - ct_2} = \frac{-1}{t_2t_3}$

$\therefore$ slope AD is t_2t_3

Equation of altitude AD is $y - \frac{c}{t_1} = t_2t_3(x - ct_1)$

or $t_1y - c = xt_1t_2t_3 - ct_1^2t_2t_3$ ...(1)

Similarly equation of altitude BE is

$t_2y - c = xt_1t_2t_3 - ct_t t_2^2t_3$ ...(2)

Solving (1) and (2), we get the orthocentre $\left(\frac{-c}{t_1t_2t_3}, -ct_1t_2t_3\right)$ which also lies on $xy=c^2$

Q. **If the normals at $(x_i, y_i), i = 1,2,3,4$ on the rectangular hyperbola, $xy = c^2$, meet at the point (α,β) then show that**

(i) $\Sigma x_i = \alpha$ (ii) $\Sigma y_i = \beta$ (iii) $\Pi x_i = \Pi y_i = -c^4$ (iv) $\Sigma x_i^2 = \alpha^2$ (v) $\Sigma y_i^{\,2} = \beta^2$

Sol. Let $(x_i, y_i) = \left(ct_i, \frac{c}{t_i}\right), i = 1,2,3,4$ are the points on the rectangular hyperbola

$xy = c^2$

Equation of normal to the hyperbola

$xy = c^2$ at $\left(ct, \frac{c}{t}\right)$ is $ct^4 - t^3x + ty - c = 0$

It passes through α, β then $ct^4 - t^3\alpha + t\beta - c = 0$

it is biquadratic equation in t. Let the roots of this equation are

t_1, t_2, t_3, t_4 then $\Sigma t_1 = \frac{\alpha}{c}$

$\Sigma t_1t_2 = 0, \Sigma t_1t_2t_3 = \frac{-\beta}{c}, t_1t_2t_3t_4 = -1$

Now

(i) $\Sigma x_i = c\Sigma t_1 = \alpha$

(ii) $\Sigma y_i = c\left(\Sigma \frac{1}{t_i}\right) = \left(\frac{\sum t_1t_2t_3}{t_1t_2t_3t_4}\right) = \beta$

(iii) $\Pi x_i = c^4\Pi t_i = -c^4$ and $\Pi y_i = c^4\left(\frac{1}{\Pi t_i}\right) = -c^4$

(iv) $\Sigma x_i^2 = c^2\left(\Sigma t_i^2\right) = c^2\left(\left(\Sigma t_1\right)^2 - 2\Sigma t_1t_2\right) = \alpha^2$

(iv) $\Sigma y_i^2 = \left(\left(\Sigma y_1\right)^2 - \Sigma y_1y_2\right) = b^2 - 2c^2\Sigma\left(\frac{1}{t_t t_2}\right) = b^2 - 2c \cdot \frac{\Sigma t_1t_2}{t_1t_2t_3t_4} = \beta^2$

Q. **The normal at one extremity of latus rectum (in 1st quadrant) of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, $a > b > 0$ meets the rectangular hyperbola $xy = 9$ at points P and Q, then**

(A) If P is $\left(6,\frac{3}{2}\right)$ $\Rightarrow$ Q is $\left(-\frac{3\sqrt{2}}{2},-3\sqrt{2}\right)$

(B) Eccentricity of hyperbola is $\sqrt{2}$

(C) If P is $\left(6,\frac{3}{2}\right)$ $\Rightarrow$ Q is $\left(-\frac{3e}{2},-\frac{6}{e}\right)$ where e is eccentricity of the given ellipse

(D) If O is origin, then product of slopes of OP and OQ is positive

Sol. **(BCD)**

Equation of normal at $\left(ae,\frac{b^2}{a}\right)$; $\frac{a}{e}x - ay = a^2e^2$

$x - ey = ae^3$

Put $y = \frac{9}{x}, x - 9\frac{e}{x} = ae^3$

$\Rightarrow x^2 - ae^3x - 9e = 0$ $\begin{matrix} \nearrow x_1 \\ \searrow x_2 \end{matrix}$

if $x_1 = 6$, $x_1x_2 = -9e$,

$x_2 = -\frac{3e}{2}$; $y_2 = \frac{9}{x_2} = -\frac{6}{e}$

$\Rightarrow$ (slope of OP) (slope of QQ) $= \left(\frac{y_1}{x_1}\right)\left(\frac{y_2}{x_2}\right) = \frac{9}{x_1^2}\frac{9}{x_2^2} = \frac{81}{x_1^2x_2^2} > 0$

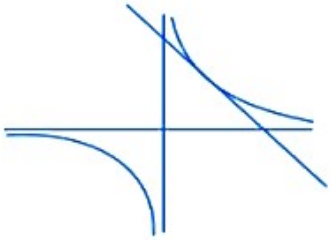
Q. **Straight line $Ax + By + D = 0$ would be tangent to $xy = c^2$, if**
(A) $A > 0$, $B > 0$ (B) $A < 0$, $B < 0$ (C) $A > 0$, $B < 0$ (D) $A < 0$, $B > 0$

Sol. **(AB)**

$y = \frac{c^2}{x}$ is decreasing function

$\Rightarrow$ slope $= -\frac{A}{B} < 0 \Rightarrow \frac{A}{B} > 0$

$\Rightarrow$ $A > 0, B > 0$ or $A < 0, B < 0$



Q. **If the normal to the rectangular hyperbola $x^2 - y^2 = 4$ at a point P meets the coordinates axes in Q and R and O is the center of the hyperbola. Then**
(A) OP = PQ **(B) OP = PR** **(C) PQ = PR** **(D) QR = 2 OP**

Sol. **(ABCD)**

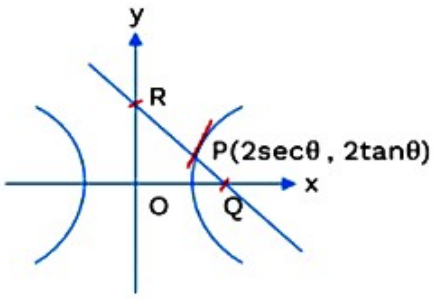
Equation of normal at P

$$\frac{x}{\sec\theta} + \frac{y}{\tan\theta} = 4$$

$\Rightarrow$ R $\equiv$ (0, 4 tan θ), Q $\equiv$ (4 secθ, 0)

$\therefore$ OP = PQ = PR = $2\sqrt{\sec^2\theta + \tan^2\theta}$

QR = $4\sqrt{\sec^2\theta + \tan^2\theta}$



Q. **The locus of a point, from where tangents to the rectangular hyperbola $x^2 - y^2 = a^2$ contain an angle of 45°, is**

(A) $\left(x^2+y^2\right)^2 + a^2\left(x^2-y^2\right) = 4a^2$ **(B)** $2\left(x^2+y^2\right)^2 + 4a^2\left(x^2-y^2\right) = 4a^2$

(C) $\left(x^2+y^2\right)^2 + 4a^2\left(x^2-y^2\right) = 4a^2$ **(D)** $\left(x^2+y^2\right)^2 + a^2\left(x^2-y^2\right) = a^4$

Sol. **(C)**

Let $y = mx \pm \sqrt{m^2a^2 - a^2}$ be two tangents and passes through (h, k). Then

$(k - mh)^2 = m^2a^2 - a^2$

$\Rightarrow m^2\left(h^2 - a^2\right) - 2khm + k^2 + a^2 = 0$

$$\Rightarrow m_1 + m_2 = -\frac{2kh}{h^2 - a^2}$$

and $$m_1m_2 = \frac{k^2 + a^2}{h^2 - a^2}$$

Now, $$\tan 45^\circ = \frac{m_1 - m_2}{1 + m_1m_2}$$

$$\Rightarrow 1 = \frac{\left(m_1 + m_2\right)^2 - 4m_1m_2}{\left(1 + m_1m_2\right)^2}$$

$$\Rightarrow \left(1 + \frac{k^2 + a^2}{h^2 - a^2}\right)^2 = \left(\frac{2kh}{h^2 - a^2}\right)^2 - 4\left(\frac{k^2 + a^2}{h^2 - a^2}\right)$$

$$\Rightarrow \left(h^2 + k^2\right)^2 = 4h^2k^2 - 4\left(k^2 + a^2\right)\left(h^2 - a^2\right)$$

$$\Rightarrow \left(x^2+y^2\right)^2 = 4\left(a^2y^2 - a^2x^2 + a^4\right)$$

$$\Rightarrow \left(x^2+y^2\right)^2 + 4a^2\left(x^2-y^2\right) = 4a^2$$

Q. **If a chord joining the points P(a secθ, a tanθ) and Q(a secϕ, a tanϕ) on the hyperbola $x^2-y^2=a^2$ is a normal to it at P, then show that $\tan\phi = \tan\theta\,(4\sec^2\theta-1)$**

Sol. Slope $= -\dfrac{\tan\theta}{\sec\theta}$

$$\frac{-\tan\theta}{\sec\theta} = \frac{\tan\theta - \tan\phi}{\sec\theta - \sec\phi}$$

$$-\tan\theta\sec\theta + \tan\theta\sec\phi = \tan\theta\sec\theta - \tan\phi\sec\theta$$

$$\tan\theta\sqrt{1+\tan^2\phi} + \tan\phi\sec\theta = 2\tan\theta\sec\theta$$

$$\tan\theta\sqrt{1+\tan^2\phi} = 2\sec\theta\tan\theta - \tan\phi\sec\theta$$

Squaring

$$\tan^2\theta\left(1+\tan^2\phi\right) = 4\sec^2\theta\tan^2\theta + \tan^2\phi\sec^2\theta + 4\tan\theta\sec^2\theta\tan\phi$$

$$\tan^2\theta + \tan^2\phi\left(\tan^2\theta - \sec^2\theta\right) = 4\sec^2\theta\tan\theta(\tan\theta - \tan\alpha)$$

$$\tan^2\theta - \tan^2\phi = 4\sec^2\theta\tan\theta(\tan\theta - \tan\alpha)$$

$$\tan\theta + \tan\phi = 4\sec^2\theta\tan\theta$$

$$\tan\phi = \tan\theta\left(4\sec^2\theta - 1\right)$$

Q. **From points on the circle $x^2+y^2=a^2$ tangents are drawn to the hyperbola $x^2-y^2=a^2$, prove that the locus of the middle points of the chords of contact is the curve $\left(x^2-y^2\right)^2 = a^2\left(x^2+y^2\right)$**

Sol. Let mid point (a, k) Point on circle is (a cos θ, 2 sin θ)
Equation of chord of contact $x\cos\theta - y\sin\theta = a$ ...(1)

Equation of chord as per (h, k)
$T = S_1$
$xh - yk = h^2 - k^2$

(1) and (2) are equation of same line

$$\frac{h}{\cos\theta} = \frac{-k}{-\sin\theta} = \frac{h^2 - k^2}{a}$$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\frac{a^2h^2}{\left(h^2 - k^2\right)^2} + \frac{k^2a^2}{\left(h^2 - k^2\right)^2} = 1$$

$$a^2\left(h^2 + k^2\right) = \left(h^2 - k^2\right)^2$$

Q. **Consider hyperbola x y=16 to find the following:**

(i) **Coordinates of vertices**
(ii) **Length of transverse axis**
(iii) **Coordinates of foci**
(iv) **Length of latus rectum**
(v) **Equations of two directrices**
(vi) **Equation of tangent at point (2,8)**
(vii) **Equation of chord which gets bisected at point (5,6)**
(viii) **Equation of tangent having slope −2**
(ix) **Equation of normal having slope 2**

Sol. We have hyperbola xy = 16.

(i) Clearly, transverse axis is x-y=0 and conjugate axis is x + y = 0
Vertices are points of intersection of transverse axis x − y = 0 and hyperbola.
So, vertices are A(4,4) and A′(-4,-4)

(ii) Length of transverse axis is $AA' = 8\sqrt{2}(= 2a)$

(iii) Foci lie at distance 'ae' from centre on line x − y = 0
Now, $ae = 4\sqrt{2}\sqrt{2} = 8$

Therefore, foci are $F_1\left(4\sqrt{2}, 4\sqrt{2}\right)$ and $F_2\left(-4\sqrt{2}, -4\sqrt{2}\right)$

(iv) Length of latus rectum $= \frac{2b^2}{a} = 2a = 8\sqrt{2}$

(as a = b for rectangular hyperbola)

(v) The directrices lies at distance $= \frac{a}{e}(= 4)$ from centre and is parallel to the conjugate axis $x + y = 0$

So, equation of two directrices are $x + y = \pm 4\sqrt{2}$

(vi) Equation of tangent at point (2,8) is T = 0

$\Rightarrow \quad \frac{2y + 8x}{2} - 16 = 0$

$\Rightarrow \quad 4x + y - 16 = 0$

Alternatively, we can differentiate the equation of to get the slope of tangent. Then, we get $y + x\frac{dy}{dx} = 0$

$\therefore \quad \frac{dy}{dx} = -\frac{y}{x}$

$\therefore \quad \left(\frac{dr}{dx}\right)_{12.8} = -\frac{8}{2} = -4$

So, equation of tangent at (2,8) is $y - 8 = -4(x - 2)$

$\Rightarrow \quad 4x + y - 16 = 0$

(vii) Equation of chord which gets bisected at point (5,6) is $T = S_1$

$\Rightarrow \quad \frac{5y + 6x}{2} - 16 = (5)(6) - 16$

$\Rightarrow 6x + 5y = 60$

(viii) To find the equation of tangent having slope −2, differentiate the curve and compare derivative to −2. Then, we get

$\therefore \quad \frac{dy}{dx} = -\frac{y}{x} = -2$

$\therefore y = 2x$

Solving this with hyperbola, we get points $P(\sqrt{8}, 2\sqrt{4})$ and $Q(-\sqrt{8}, -2\sqrt{8})$ on the hyperbola where slope. tangent is −2

Equation of tangent at point P is $y - 2\sqrt{8} = -2(x - \sqrt{8})$

$\Rightarrow \quad 2x + y = 4\sqrt{8}$

(ix) We have $\frac{dy}{dx} = -\frac{y}{x}$

Thus, slope of normal at any point on the curve is $-\frac{dx}{dy} = \frac{x}{y} = 2$ (given)

$\therefore x = 2y$

Solving this with hyperbola, we get points $A(2\sqrt{8}, \sqrt{8})$ of and $B(-2\sqrt{8}, -\sqrt{8})$ on the hyperbola where normal is 2.

Equation of normal at point A is $y - \sqrt{8} = 2(x - 2\sqrt{8})$

$\Rightarrow \quad 2x - y = 3\sqrt{8}$

Equation of normal at point B is $y + \sqrt{8} = 2(x + 2\sqrt{8})$

$\Rightarrow \quad 2x - y = -3\sqrt{8}$

Q. **If a circle and the rectangular hyperbola $xy = c^2$ meet in the four points t_1, t_2, t_3 & t_4 then**

(a) $t_1 t_2 t_3 t_4 = 1$

(b) **the centre of the mean position of the four points bisects the distance between the centres of the two curves.**

(c) **the centre of the circle through the points t_1, t_2 & t_3 is:**

$$\left\{\frac{c}{2}\left(t_1 + t_2 + t_3 + \frac{1}{t_1 t_2 t_3}\right), \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + t_1 t_2 t_3\right)\right\}$$

Sol. **(a)** Let the equation of the circle be $x^2 + y^2 + 2gx + 2fy + d = 0$ ...(i)

Solving Eq. (i) and the equation of hyperbola, we have

$$x^2 + \frac{c^4}{x^2} + 2gx + 2f\frac{c^2}{x} + d = 0$$

$$\Rightarrow \quad x^4 + 2gx^3 + dx^2 + 2fc^2x + c^4 = 0 \qquad \text{...(ii)}$$

From Equation (i)

$x_1 x_2 x_3 x_4 = c^4$

$c^4 \left[t_1 t_2 t_3 t_4\right] = c^4$

$\Rightarrow \quad t_1 t_2 t_3 t_4 = 1$

(b) Again, centre of the mean position of the four points of intersection is

$$\left(\frac{\sum x_i}{4}, \frac{\sum y_i}{4}\right)$$

Now from Eq. (i),

$x_1 + x_2 + x_3 + x_4 = -2g$

$$\Rightarrow \quad \frac{\sum x_t}{4} = -\frac{g}{2}$$

using $xy = c^2$ we have

$$y_1 + y_2 + y_3 + y_4 = c^2\left[\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \frac{1}{x_4}\right]$$

$$= \frac{c^2}{x_1x_2x_3x_4}\sum x_1x_2x_3$$

$$= \frac{c^2}{c^4}\left(-2fc^2\right) = -2f$$

$$\Rightarrow \quad \frac{\sum y_i}{4} = -\frac{f}{2}$$

Hence $\left(\frac{\sum x_i}{4}, \frac{\sum y_i}{4}\right) = \left(-\frac{g}{2}, -\frac{f}{2}\right)$

(c) Again from eq. (6), we have $t_4 = \frac{1}{t_1t_2t_3}$, so we get

$$-g = \frac{c}{2}\left(t_1 + t_2 + t_3 + \frac{1}{t_1t_2t_3}\right) \text{ and}$$

$$-f = \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + t_1t_2t_3\right)$$

The coordinates of the centre of the circle are therefore

$$\left(\frac{c}{2}\left(t_1 + t_2 + t_3 + \frac{1}{t_1t_2t_3}\right), \frac{c}{2}\left(\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + t_1t_2t_3\right)\right)$$

Q. **A circle of radius 'r' cuts a rectangular hyperbola in four points P,Q,R and S, and C is the centre of the hyperbola, prove that $CP^2 + CQ^2 + CR^2 + CS^2 = 4r^2$**

Sol. Let $x^2 + y^2 + 2gx + 2fy + k = 0$ be the circle and $xy = c^2$ be the hyperbola.

Let a point $\left(ct, \frac{c}{t}\right)$ is common point of two curves, then we have

$$c^2t^2 + \frac{c^2}{t^2} + 2gct + \frac{2fc}{t} + k = 0$$

Or $c^2t^4 + 2gct^3 + kt^2 + 2fct + c^2 = 0$ ...(1)

Roots t_1, t_2, t_3, t_4 will represent the parameters of the four points of intersection,

where

$$t_1 + t_2 + t_3 + t_4 = -\frac{2g}{c} \quad \ldots(2)$$

$$t_1t_2 + t_1t_3 + t_1t_4 + t_2t_3 + t_2t_4 + t_3t_4 = \frac{k}{c^2} \quad \ldots(3)$$

$$t_2t_3t_4 + t_3t_4t_1 + t_4t_1t_2 + t_1t_2t_3 = -\frac{2f}{c} \quad \ldots(4)$$

And $t_1t_2t_3t_4 = 1$ ...(5)

From eqs. (4) and (5), we get $\frac{1}{t_1} + \frac{1}{t_2} + \frac{1}{t_3} + \frac{1}{t_4} = -\frac{2f}{c}$ ...(6)

Now $CP^2 + CQ^2 + CR^2 + CS^2$

$$= c^2\left[t_1^2 + \frac{1}{t_1^2} + t_2^2 + \frac{1}{t_2^2} + t_3^2 + \frac{1}{t_3^2} + t_4^2 + \frac{1}{t_4^2}\right]$$

$$= c^2\left[\left(\Sigma t_1\right)^2 - 2\Sigma t_1t_2 + \left(\Sigma\frac{1}{t_1}\right)^2 - 2\Sigma\frac{1}{t_1t_2}\right]$$

$$= c^2\left[\frac{4g^2}{c^2} - \frac{2k}{c^2} + \frac{4f^2}{c^2} - \frac{2k}{c^2}\right]\left[\because \Sigma\frac{1}{t_1t_2} = \frac{\Sigma t_1t_2}{t_1t_2t_3t_4} = \Sigma t_1t_2\right]$$

$$= 4\left(g^2 + f^2 - k\right)$$

$= 4r^2$, where $r = \sqrt{g^2 + f^2 - k}$ is the radius of the circle.